# Where Theorem 3.11 sits: the log-volume content of the results cited in the proof of IUT III, Corollary 3.12, does not imply its conclusion

Toshihisa Urahigashi
123026.urahigashi.toshihisa@gmail.com

Version 1.2.0 — 10 November 2026
ORCID 0009-0004-0460-6242
doi:10.5281/zenodo.24044582

**Abstract**

The inequality of [IUT3, Cor. 3.12] is obtained in Step (xi) of its proof, where the author states that he will be concerned only with certain qualitative properties of the multiradial algorithm of [IUT3, Thm. 3.11], "temporarily forgetting" its internal structure. We list every target that the corollary and its proof cite (164 citations of 63 targets; counting rule in §2, list in Appendix A), keep the 22 that the proof uses as definitions or premises, and state their content as axioms about normalized log-volumes of regions in tensor packets with Haar measure, at the strength of the source. A weak reading of the property (SHE) of [IUT3, Rem. 3.11.1(iii)] follows from these axioms. We give models of the axioms in which the conclusion of [IUT3, Cor. 3.12] fails. One of them is built from the actual local objects at a finite extension of $\mathbb{Q}_p$, ramified or not, that carries the local data of a theta function: the tensor packets of the field, the mono-analytic integral structure with the normalization of [IUT4, Prop. 1.4(i)], the indeterminacies (Ind1) and (Ind2) acting through all permutations of tensor factors and all automorphisms of the log-shells, values of the theta series of [EtTh, Prop. 1.4], a container that is a multiple of the log-shell packet and contains all horizontal and vertical images, and, as holomorphic hull, the smallest multiple of the ring of integers of the packet that contains the container. The models survive the addition of any finite set of further places, and the closed loop to which [IUT3, Rem. 3.11.1(iii)] relates (SHE) holds in them. Two strong readings of (SHE), and the equality (9-1) that the interim report of Project LANA [LANA] identifies as the missing step, read at the level of log-volumes, each imply the conclusion, and none of them follows from the axioms. The whole distance between the axioms and the conclusion therefore lies between the weak and the strong readings of (SHE). We display this position in one Venn diagram, drawing each statement as the class of structures in which it holds: the models of the cited theory lie inside the class of weak (SHE) and not inside that of the conclusion, and among them the conclusion holds exactly where strong (SHE) holds. The statement "3.11.5" that Mochizuki isolates in his report on formalization [Form, §4] corresponds to that intersection. The conclusion does follow if the images of the $\Theta$-pilot and the $q$-pilot are identified, but then the degree scale of the $\Theta$-link is 1. Negative results of this kind were proved before, for abstractions of the passage [Higuchi, Pajo] and in Lean [Takkun, lana]; the public repository lana-agents/iut of Project LANA proves non-derivability from a package collecting what it proves at the level of [IUT3, Thm. 3.11] and separates two readings of (SHE) [lana]. The results here were obtained independently of these Lean works, which were made public first. What this note adds is that the axioms are the results that the proof actually cites, at the strength of the source, and that the models are built on actual local objects. All statements proved here are checked in Lean 4 with Mathlib, with no axioms beyond the standard three. *We do not claim that [IUT3, Cor. 3.12] is false,*

*nor anything about abc. The result concerns what follows from the statements that the proof cites, including what [IUT3, Thm. 3.11] states about the output of its algorithm; whether the construction of the algorithm itself supplies the missing membership is not decided here.*

**2020 Mathematics Subject Classification.** Primary 14H25; Secondary 11G05, 11S15, 03B35.
**Keywords.** inter-universal Teichmüller theory, log-volume, log-shell, holomorphic hull, independence, formal verification.

# 1 What this is

Throughout, [S] marks a fact about a cited text, [P] a statement proved here (and checked in Lean), and [H] a reading that is not proved.

**Claimed.**

(i) [S] The statement and proof of [IUT3, Cor. 3.12] cite 63 distinct targets 164 times, by the counting rule fixed in §2; the targets are listed, with their counts and roles, in Appendix A. Of these, 22 are used as definitions or premises. The two steps (xi-e) and (xi-f), at which the inequality is drawn, invoke no numbered result as a premise: (xi-e) names the multiradial construction algorithm of [IUT3, Thm. 3.11], whose output it describes, and (xi-f) the statement of [IUT3, Cor. 3.12] itself. The properties (IPL) and (SHE) do not occur in the statement of [IUT3, Thm. 3.11]; they are introduced in [IUT3, Rem. 3.11.1(iii)]. In (xi-b) the author states that, as far as the algorithm of [IUT3, Thm. 3.11] is concerned, Step (xi) is concerned only with its properties (IPL), (SHE) and (APT) (§2).

(ii) [P] The 22 results, stated at the level of normalized log-volumes of regions in tensor packets, do not imply the conclusion of [IUT3, Cor. 3.12] (Theorem 10). A weak reading of (SHE), in which the possible images are computed from the container common to both sides, follows from them (Proposition 5).

(iii) [P] Neither of two strong readings of (SHE), nor the equality (9-1) of [LANA, §9.2], read at the level of log-volumes as in §3, nor the approximate form of the conclusion allowed in (xi-f), follows from them (Theorem 10, Remark 7). Each of these implies the conclusion (Proposition 6; for the approximate form, Remark 7). So the whole distance between the axioms and the conclusion lies between the weak and the strong readings of (SHE) (Corollary 11).

(iv) [P] This remains so after adding any finite set of places, and after enlarging the container at the bad place by a factor $211^c$ for any integer $c \geq 0$ (Theorems 14 and 15).

(v) [P] It remains so when the model is built from the actual local objects at any finite extension $L$ of $\mathbb{Q}_p$, ramified or not, that carries a theta datum (Definition 19): mono-analytic integral structure, actions of (Ind1) and (Ind2) through all automorphisms of the log-shells (both nontrivial), values of the theta series, a container that is a multiple of the log-shell packet, and the hull of [IUT3, Rem. 3.9.5(i)]; the enlargement of the hull over the image of the $\Theta$-pilot is bounded by a constant that depends only on $L$ (Theorem 22). An example is given with ramification index divisible by 98.

(vi) [P] The composite of (IPL) and the closed loop to which [IUT3, Rem. 3.11.1(iii)] relates (SHE) can be placed on every model of the axioms, and restricting the possible images, as the proof of [IUT3, Cor. 3.12] allows, can only lower the log-volume of the hull (§7).

(vii) [P] If the images of the Θ-pilot and the $q$-pilot are identified, the conclusion follows, and the degree scale of the Θ-link is then 1. Conversely, the conclusion follows whenever the degree scale is 1 and the image of the Θ-pilot has nonzero measure; so in every model above the degree scale differs from 1 (Proposition 26).

**Not claimed.** That [IUT3, Cor. 3.12] is false. That the missing statement cannot be supplied by the construction underlying [IUT3, Thm. 3.11], as opposed to its statement and the qualitative properties that Step (xi) uses. (In the terms of [TC, §6.3], the result concerns the content of the cited statements taken as a specification; in the mockup of [TC, §9], the place where the claim for every initial Θ-datum would enter is the oracle for Step (xi), whose value is reverse-computed from the conclusion.) That the axioms capture every consequence of the cited results: they capture their content at the level of log-volumes, which is the level at which Step (xi) draws its inequality. That the models come from an elliptic curve, or from an initial Θ-datum, at every place: they are built from local data, and the family in $N$ replaces the Tate parameter by its $N$-th power. That the theta values of §6 are those reconstructed by the algorithms of the source: they are values of the theta series of [EtTh, Prop. 1.4]. That the groups of (Ind1) and (Ind2) of the source are constructed: in §6 they are replaced by the full group of automorphisms of the log-shells, through which the source's groups act. That the possible images of the models are the output of the algorithm of [IUT3, Thm. 3.11]. Anything about *abc*.

**What is new.** The assertion that [IUT3, Cor. 3.12] does not follow from the statement of [IUT3, Thm. 3.11] has been made before, in several forms (§10), and several works prove formal versions of it. Two preprints do so for the same passage in formalizations that keep less of the structure of the source. Higuchi [Higuchi] proves non-derivability for a deliberately qualitative package extracted from Steps (xi-e) and (xi-f): abstract labels, a common container, abstract linkage and common-container cohabitation, an arbitrary hull operator, an arbitrary log-volume map, and a closed half-line of output values. Pajo [Pajo] proves a no-go theorem for a typed but scale-free monoid/hull/log-volume abstraction, by an active relative rescaling of the Θ-side embedding into the container. Pajo states that his abstraction is not a faithful model of IUT [Pajo, §§2, 14], and Higuchi that his purpose is not to formalize all of IUT [Higuchi, §3].

[S] In Lean, Takkun-kohinata/IUT_LEAN [Takkun] (July 2026) proves, for its transcription of the passage, that the volume bound of the multiradial representation together with the containment is equivalent to the height inequality (`thetaHullTranscription_iff`); that strict identification of the copies, with volume-preserving indeterminacies and a hull no larger than the Θ-pilot, is contradictory (`strict_collapse`); and that the slack between the hull and the Θ-pilot cannot come from the orbits of (Ind1) and (Ind2), only from the hull (`hull_exceeds_orbit`), and is at least $(\text{factor}-1)|\log q|$ (`slack_lower_bound`). Saito [Saito] (July 2026) audits the passage from (xi-a)–(xi-e) to the membership of (xi-f) against a registered inventory of the source, separates several membership statements, and proves in Lean that every registered derivation of the membership uses a node of a target seed; its source-level conclusion is conditional on a faithfulness hypothesis that it leaves open. sfingali [sfingali] (August 2026) formalizes in Lean the core of the objection of [SS]: the monodromy of the identification, the empty inequality under a consistent identification, and (9-1) as two identifications of pointed real lines. Proposition 26, Corollary 11 and the remark after it state results of the same shape for the cited theory.

For the same passage, the public repository lana-agents/iut of Project LANA [lana] collects what the repository proves at the level of [IUT3, Thm. 3.11] into one package (`Thm311Package`), decouples the numbers ($|\log q|$ and the volume of the container) from the data, and proves in Lean that Step (xi) cannot be derived from the fields of the package. The same file shows that Step (xi) adds to the package exactly the Scholze–Stix inequality (`xi_iff`; in our notation $(s^{-1}-1)\,a \le V$,

where $a = |\log q|$, $V$ is the volume of the container, and $s^{-1} = \ell(\ell+1)/12$ is the inverse of our degree scale $s$), and that Step (xi) for all initial $\Theta$-data gives *abc*. The same repository separates two readings of (SHE) in `Iut/Wall/Constraints.lean` (12 October 2026): (C2), that the output is a function of the Hodge theater on each side, and (C3), that some output possibility has the volume class of the $q$-pilot region, which it proves equivalent to the wall (`sheq_iff_wall`); and it reads (9-1) for one choice as an equation of log-volumes (`etaAnab_eq_etaQ_iff`, `Iut/Wall/EtaAlg.lean`, 11 October 2026). (C2) corresponds to our weak reading of (SHE); (C3), and (9-1) read as an equation of log-volumes, correspond to our (9-1); and the wall corresponds to the value reading (Definition 4). In addition, Definition 4 separates the membership of the $q$-pilot region itself (the carrier reading) from that of its volume class. The results of the present note were obtained independently of these files (its text and Lean sources were completed in September 2026), and the repository was made public first. Table 1 records how the formalizations are set up: our axioms are taken from an inventory of what the proof cites, and our models are built on actual local objects.

The point of the present note is not to repeat these results, but to test whether the obstruction disappears once the numerical structure that the cited results do state is restored. It does not. The axioms are taken from an inventory of what the proof actually cites, at the strength of the source; the log-volume is the Haar log-volume of [IUT3, Prop. 3.9] on the common tensor packets, so both pilots are measured by the same function; the degree scale is not chosen but forced by the two volumes; the hull is the smallest region of the form $\lambda \cdot \widetilde{\mathcal{O}}$, with $\widetilde{\mathcal{O}}$ the ring of integers of the packet, as in [IUT3, Rem. 3.9.5(i)]; the models use actual $p$-adic carriers, so that the log-volumes are actual numbers; at a ramified place the integral structure, the values of the theta series and the hull are built from actual local objects, the container is a multiple of the actual log-shell packet that contains all horizontal and vertical images, and the indeterminacies act through the full automorphism groups of the log-shells; and the failure is robust in the number of places. What still does not follow is the membership required in (xi-f), that is, strong (SHE), while weak (SHE) does follow (Corollary 11); and the dependence of the conclusion on the degree scale, and on the identification of the two pilots, is a theorem. Table 1 compares the four formalizations. The obstruction found in the earlier papers is therefore not merely an artifact of forgetting the numerical structure. We still do not claim that [IUT3, Cor. 3.12] is false, nor that the models come from initial $\Theta$-data.

The results are also checked differently. Everything proved here is checked in Lean 4 with Mathlib, including that the models satisfy every axiom of Definition 2, that the conclusion fails in them, that weak (SHE) follows and the strong readings do not, and that identifying the two pilots, or a degree scale 1 together with an image of the $\Theta$-pilot of nonzero measure, gives the conclusion. The Lean proofs use no axioms beyond the three standard ones of Lean, and mutations of the models break the declarations they target (§11). Lean checks that these statements follow from the definitions and axioms of this note; it does not check that the axioms state the cited results correctly. That correspondence rests on the inventory of Table 2 and on the source attached to each axiom of Definition 2, whose strength is discussed in Remark 3; it is checked against the source, not by Lean.

### Statements as classes of structures

The position described by Corollary 11 is seen at a glance when each statement is drawn as the class of structures in which it holds (Figure 1). A structure is, in the language of §3, a packet system together with possible images, a container, a hull and a $q$-pilot. The stronger a statement, the smaller its class, and an implication $P \Rightarrow Q$ is the inclusion of the class of $P$ in that of $Q$.

| | Higuchi [Higuchi] | Pajo [Pajo] | lana-agents [lana] | this note |
|---|---|---|---|---|
| formalized | a qualitative package from Steps (xi-e), (xi-f) | a typed, scale-free monoid/hull/log-volume abstraction | a package collecting what the repository proves at the level of Thm. 3.11 (`Thm311Package`) | the log-volume content of the 22 results that the proof cites |
| log-volume | an arbitrary map $r\colon C \to \mathbb{R}$ | a map to each typed domain, and separate evaluations $F_q$, $F_\Theta$ | the numbers $a = \lvert \log q \rvert$, $V$ (container) and $R$ (hull), decoupled from the data | one Haar log-volume on the common packets |
| degree scale | none | free; the countermodel rescales the $\Theta$-side | fixed by $\ell$ (our $s^{-1} = \ell(\ell+1)/12$) | forced by the two volumes (3/14 at one place) |
| hull | any extensive, monotone, idempotent operator | an abstract map on each side | the hull of the $\Theta$-pilot region, of log-volume $R$; outputs have volume at most $R$ | the smallest $\lambda \cdot \widetilde{\mathcal{O}}$ containing the possible images |
| container | a set with a map to the labels | a set with embeddings of both sides and a relation | IUT's container; the (Ind3)-choices have volumes exactly $(-\infty, V]$ | a multiple of the log-shell packet |
| carriers | none | none | none (a numerical family); packages from genuine initial $\Theta$-data are constructed separately (`ofThetaData`) | $\mathbb{Q}_{211}$; finite extensions of $\mathbb{Q}_p$; $\mathbb{Q}_5$ and $\mathbb{R}$ as extra places |
| (Ind1)–(Ind3) | not modelled | not modelled | unit-induced, volume-preserving Kummer identifications (fields); (Ind3) through the container | acting on the packets; (Ind3) through the container |
| result | the package does not give the inclusion of (xi-f) | the container comparison does not give the numerical inequality | Step (xi) $\iff (s^{-1}-1)\,a \le V$ (`xi_iff`); the fields alone do not give Step (xi); Step (xi) for all data gives *abc* | the cited content gives weak (SHE) but not strong (SHE) |
| relation to the source | not a formalization of all of IUT (§3) | not a faithful model of IUT (§§2, 14) | that the fields exhaust what the repository proves is stated as audited, not a theorem | each axiom tagged with its source (Def. 2, Rem. 3); the models do not come from initial $\Theta$-data |
| checked by | written proof (§5) | written proof (§8) | Lean | written proof, and Lean 4 with Mathlib (§11) |
| claims that Cor. 3.12 is false | no | no | no | no |

Table 1: Four formalizations of the passage from [IUT3, Thm. 3.11] to [IUT3, Cor. 3.12], compared; other earlier formal work is described in §1. The column lana-agents is a result obtained independently of this note and made public before it. The rows record what each formalization keeps and how its results are checked; they do not assert that one theory contains another.

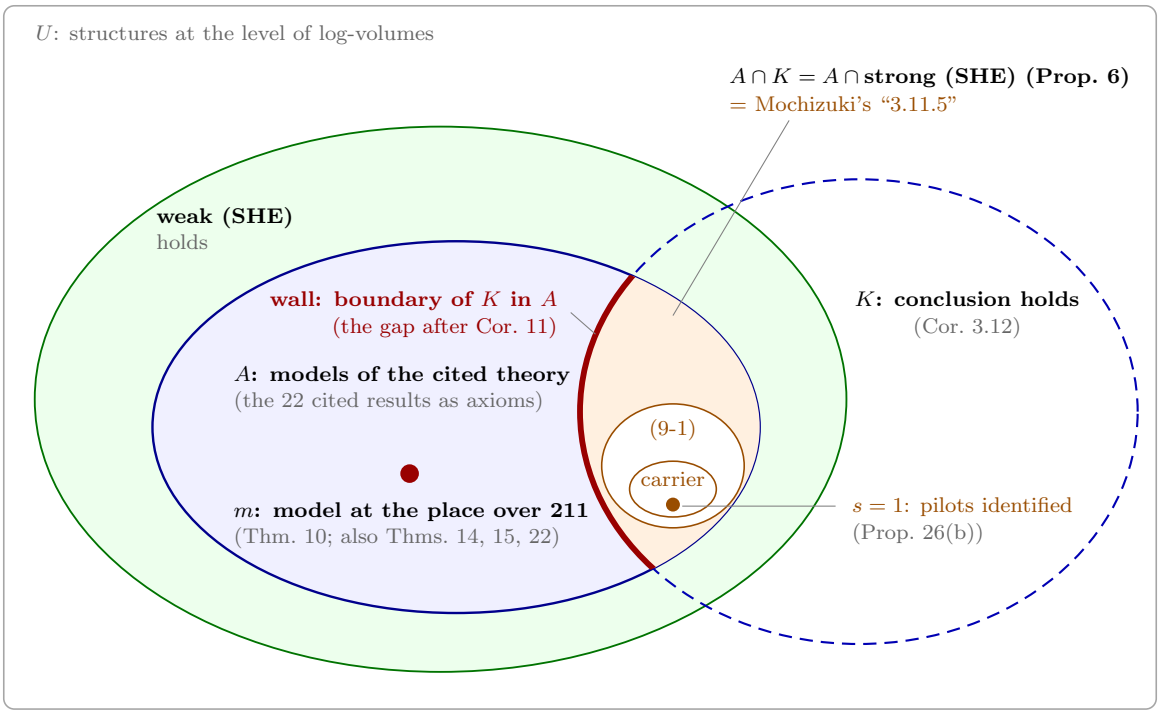



Figure 1: Each statement drawn as the class of structures in which it holds; the stronger the statement, the smaller the class. Sizes and shapes are schematic, and the positions of the points measure nothing. The exact relations are (a)–(e) below; no inclusion is claimed that is not listed there. The reconstruction content of Theorem 3.11 (Kummer classes of theta values, cyclotomic rigidity) does not appear at this level: Step (xi-b) forgets it.

(a) [P] $A$ lies inside the class of weak (SHE): weak (SHE) holds in every model of the cited theory (Proposition 5).

(b) [P] $A$ does not lie inside $K$: $A$ contains a model $m$ in which the conclusion fails (Theorem 10); the same holds after adding places, after enlarging the container, and when the model is built on actual local objects (Theorems 14, 15 and 22).

(c) [P] Inside $A$, the class $K$ coincides with the class of strong (SHE) in the value reading; the class of the carrier reading lies inside that of (9-1), and that of (9-1) inside $K$ (Proposition 6).

(d) [P] A structure in which the images of the two pilots are identified satisfies the carrier reading and lies in $K$, and its degree scale is then $s = 1$; conversely, $s = 1$ together with an image of the $\Theta$-pilot of nonzero measure gives $K$ (Proposition 26(b)(c)).

(e) [H] The statement "3.11.5" (Theorem 3.11 together with Remark 3.9.5) that Mochizuki isolates in his report on formalization [Form, §4] corresponds to the intersection of $A$ with strong (SHE), that is, to $A \cap K$, and implies the conclusion (§10). Whether a model in $A$ crosses the boundary of $K$ is whether the gap $\mathrm{vol}(q) - \mathrm{vol}(H)$ of the remark following Corollary 11 is covered; this is the position of the inequality in `xi_iff` of [lana].

[S] In Step (xi-b) the author states that he takes the point of view of "temporarily forgetting" that the multiradial construction algorithm of [IUT3, Thm. 3.11] involves $\Theta$-pilot objects, theta

functions/values and mono-theta environments (§2). [H] Hence the reconstruction content of Theorem 3.11 does not appear at the level of Figure 1: within Step (xi), strengthening it does not push the model $m$ out of $A$, since Step (xi) reads only (IPL), (SHE) and (APT). This does not exclude that the construction itself supplies the missing statement (see *Not claimed* above).

## 2 The step

[S] [IUT3, Cor. 3.12] asserts that $-|\log(\underline{\underline{\Theta}})| \geq -|\log(\underline{\underline{q}})|$, where $-|\log(\underline{\underline{q}})|$ is the log-volume of the $q$-pilot object and $-|\log(\underline{\underline{\Theta}})|$ is the procession-normalized log-volume of the holomorphic hull of the union of the possible images of the $\Theta$-pilot object, subject to the indeterminacies (Ind1), (Ind2), (Ind3). The inequality is obtained in Step (xi) of the proof; the last step, (xii), comments on the role of the global realified Frobenioids (§9). In (xi-b) the author writes (here and below, emphasis in the original is omitted):

> Also, in this context, we observe that, although the multiradial construction algorithm of Theorem 3.11 in fact involves the $\Theta$-pilot object at $(0, 0)$, in the present discussion of Step (xi), we shall only be concerned with qualitative logical aspects/consequences of this construction algorithm, i.e., with the
>
> - input prime-strip link (IPL),
> - simultaneous holomorphic expressibility (SHE), and
> - algorithmic parallel transport (APT)
>
> properties discussed in Remark 3.11.1, (iii), (iv), (v). That is to say, we shall take the point of view of "temporarily forgetting" — cf. the discussion of hidden internal structures (HIS) in Remark 3.11.1, (iv) — the fact that the multiradial construction algorithm of Theorem 3.11 in fact involves $\Theta$-pilot objects, theta functions/values, mono-theta environments. Alternatively, in the discussion to follow, we shall, roughly speaking, think of the multiradial construction algorithm of Theorem 3.11 as "some" algorithm that transforms a certain type of input data into a certain type of output data and, moreover, satisfies certain properties (IPL) and (SHE).

Step (xi-f) reads, in part:

> Thus, we conclude from (xi-e) that the construction of the subset $\mathbb{R}_{\leq -|\log(\underline{\underline{\Theta}})|} \subseteq \mathbb{R}$ of possible pilot-object log-volumes of output data is subject to the condition that this construction of output data possibilities constitutes, in particular, a construction [perhaps only up to some sort of "approximation", as a result of various indeterminacies] of the pilot-object log-volume of the input data ($\mathcal{F}^{\vdash\blacktriangleright\times\mu}$-)prime-strip, namely, $-|\log(\underline{\underline{q}})| \in \mathbb{R}$.

The inclusion $-|\log(\underline{\underline{q}})| \in \mathbb{R}_{\leq -|\log(\underline{\underline{\Theta}})|}$ is then the conclusion.

[S] We read every citation in the statement and the proof of [IUT3, Cor. 3.12] and classified its role (Table 2; full list in Appendix A). The statement, which cites the definitions of the two log-volumes, is counted together with the proof. Counting rule. An *occurrence* is one textual naming of a target in the English original (author version, pp. 173–186, from the heading of Corollary 3.12 to just before Remark 3.12.1): a list counts once per target named, re-naming the same target inside one bracket counts again, citations inside Figure 3.6 count, and forward references count. References to Corollary 3.12 itself are excluded. A *target* is a (document, kind, number) triple; whole works, sections and figures are targets in their own right, and a section range such as [IUT1,

| role in the proof | targets | examples |
|---|---|---|
| definition (A) | 5 | all: [IUT1, Def. 3.1], [IUT2, Cor. 4.10]; Def. 2.4, Rem. 3.1.1, Def. 3.8 |
| definition and premise (A+B) | 2 | all: Prop. 3.9, Rem. 3.9.5 |
| premise (B) | 15 | all: Thm. 1.5, Thm. 3.11; Rems. 1.2.2, 1.5.2, 1.5.4, 2.4.2, 3.8.1, 3.9.2, 3.9.3, 3.9.6, 3.9.7, 3.10.1, 3.11.1, 3.11.3, 3.12.2 |
| background | 33 | e.g. [EtTh], [IUT2, §2, §3], [IUT1, §§4, 5, 6], [AbsTopIII], Prop. 3.2, [AbsAnab], Prop. 1.2.1 |
| comments and figures | 8 | Rems. 2.1.1, 3.6.2, [IUT2, Rem. 3.6.4], [IUT4], [IUT4, Rem. 2.3.2]; Figs. 3.6–3.8 |

Table 2: The 63 targets cited in the statement and proof of [IUT3, Cor. 3.12], by role. Kept: A, A+B and B, 22 targets. Full list with counts: Appendix A.

§§4, 5, 6] counts as the three sections named. This gives 164 occurrences of 63 targets. (Versions 1.0.0–1.1.1 reported 152/61 from an earlier inventory extracted from a Japanese translation, which grouped [IUT1, §§4, 5, 6] as one target and missed twelve textual occurrences; the set of targets is otherwise the same, and no role assignment changes. See "Changes in version 1.2.0" below.) Unless otherwise indicated, numbered items in the table are those of [IUT3]. Three sources used below are *not* cited in Corollary 3.12 directly ($^\dagger$): [IUT3, Prop. 3.5(ii)]$^\dagger$, used for (C3), enters through [IUT3, Thm. 3.11(ii)], in the passage quoted in Remark 3; [IUT3, Prop. 3.2(ii)]$^\dagger$, used in Definition 1(i) for the subquotients, enters through the situation of [IUT3, Prop. 3.9(i)] and through [IUT3, Thm. 3.11(i)(a)]; and [IUT3, Rem. 3.9.1]$^\dagger$, used in Definition 1 only to fix conventions (the ordering of the branches), enters with [IUT3, Prop. 3.9]. No other statement is used below without occurring in the range counted above. Theorem 3.11 is stated without (IPL) and (SHE); the two properties are introduced in [IUT3, Rem. 3.11.1(iii)] ("IPL" is mentioned before, in a forward reference in [IUT3, Rem. 3.9.5(ix)]), where (SHE) is said to correspond to the closed loop of [IUT3, Rem. 3.9.5(viii)(ix)] (§7). [S] Steps (xi-e) and (xi-f) invoke no numbered result as a premise: (xi-e) names the multiradial construction algorithm of Theorem 3.11, whose output it describes, and (xi-f) the statement of Corollary 3.12.

[H] The passage from (xi-b) fixes the level of abstraction of this note. The axioms below record what the cited results say about log-volumes, and the readings of (SHE) record what Step (xi) may take from that property; the internal structure that (xi-b) sets aside is not used.

## 3 The cited results as axioms

**Definition 1** (packets)**.** A *packet system* consists of $\ell^* \geq 1$; a finite set of places $v$; for each place $v$ a finite set $V_v$ (the direct summands) and, for each $\beta \in V_v$, two groups $A_{v,\beta}$ (automorphisms of a component of a prime-strip) and $\mathrm{Ism}_{v,\beta}$, the elements of the latter having order dividing 2 when $v$ is archimedean; and for each place $v$ and stage $j \in \{0, \ldots, \ell^*\}$: a locally compact abelian topological group $P_{v,j}$ with a regular Haar measure $\mu_{v,j}$; a compact subset $\mathcal{O}_{v,j}$ with nonempty interior (the mono-analytic integral structure [IUT3, Prop. 3.2(ii), Rem. 3.9.1(i)]); a weight $w_{v,j} > 0$ and a constant $a_{v,j}$ (accounting for the discrepancy between the mono-analytic and the local holomorphic integral structures discussed in [IUT3, Rem. 3.9.1]); and three actions on $P_{v,j}$ by automorphisms of topological groups preserving $\mathcal{O}_{v,j}$, each respecting identity and composition: of the permutations

of the labels $\{0,\dots,j\}$, of the families $(g_{\alpha,\beta})$ with $g_{\alpha,\beta}\in A_{v,\beta}$ indexed by labels $\alpha$ and summands $\beta$, and of the families of elements of the groups $\mathrm{Ism}_{v,\beta}$ indexed in the same way. (Ind1) sends a region to its image under a family of automorphisms of prime-strips composed with a permutation of the labels at each stage, the permutation being the same at all places; (Ind2) sends it to its image under a family of copies of Ism. A *region* is a family $x=(x_{v,j})$ of subsets $x_{v,j}\subseteq P_{v,j}$; it has *nonzero measure* if $\mu_{v,j}(x_{v,j})\neq 0$ for all $v$ and $j$. Its *normalized log-volume* is

$$\mathrm{vol}(x)=\sum_{v}\frac{1}{\ell^{*}}\sum_{j=1}^{\ell^{*}}\Big(w_{v,j}\big(\log\mu_{v,j}(x_{v,j})-\log\mu_{v,j}(\mathcal{O}_{v,j})\big)+a_{v,j}\Big).$$

We write $x\le y$ if $x_{v,j}\subseteq y_{v,j}$ for all $v,j$.

This is [IUT3, Prop. 3.9(i)] with the procession normalization of [IUT3, Rem. 3.1.1]: an average over the labels $j\in\mathbb{F}_{\ell}^{*}$ and a sum over places. By [IUT3, Prop. 3.9(iii)] it computes arithmetic degrees. Because the automorphisms preserve $\mathcal{O}_{v,j}$, they preserve the Haar measure, and hence vol.

**Definition 2** (cited theory)**.** A *cited theory* is a packet system with regions $\theta$ (image of the $\Theta$-pilot), $q$ (image of the $q$-pilot), $C$ (the container), $H$ (the hull), a set $\mathcal{P}$ of regions (the possible images), a class $\Lambda$ of subsets of the $P_{v,j}$ (in the source, the sets $\lambda\cdot\widetilde{\mathcal{O}}$ for the ring of integers $\widetilde{\mathcal{O}}$ of the packet; see Remark 3), and a real number $s$ (the degree scale), such that:

(C1) $\theta\in\mathcal{P}$. [IUT3, Def. 3.8(i), Thm. 3.11(i)]

(C2) $\mathcal{P}$ is closed under (Ind1) and (Ind2) of Definition 1. [IUT3, Thm. 3.11(i)]

(C3) Every $x\in\mathcal{P}$ satisfies $x\le C\le\mathcal{O}$; and if $x\in\mathcal{P}$ and $x\le y\le C$ then $y\in\mathcal{P}$. (Ind3), [IUT3, Thm. 3.11(ii), Prop. 3.5(ii)]

(C4) $H_{v,j}\in\Lambda$ for all $v,j$; $x\le H$ for all $x\in\mathcal{P}$; $H\le T$ for every region $T$ with all $T_{v,j}\in\Lambda$ and $x\le T$ for all $x\in\mathcal{P}$; and $H\le\mathcal{O}$. [IUT3, Rem. 3.9.5(i)]

(C5) $q\le\mathcal{O}$, every $\mu_{v,j}(q_{v,j})\neq 0$, and $\mathrm{vol}(q)<0$. [IUT1, Def. 3.1(b)]

(C6) $s>0$ and $\mathrm{vol}(q)=s\cdot\mathrm{vol}(\theta)$. [IUT3, Def. 3.8(ii)]; see Remark 3

(C7) For every $x\in\mathcal{P}$ there is $c>0$ with $\mathrm{vol}(q)=c\cdot\mathrm{vol}(x)$. (IPL), [IUT3, Rem. 3.11.1(iii)]

Put $-|\log(\underline{\underline{\Theta}})|:=\mathrm{vol}(H)$ and $-|\log(\underline{\underline{q}})|:=\mathrm{vol}(q)$. The *conclusion* is $\mathrm{vol}(q)\le\mathrm{vol}(H)$.

*Remark* 3 (on the strength of the axioms). [S] (C3): [IUT3, Thm. 3.11(ii)] imposes the upper semi-compatibility (Ind3) on the Kummer isomorphisms of tensor packets of log-shells. For the splitting monoids (theta values) and the copies of $F_{\mathrm{mod}}$ it states instead that they are mutually compatible along the log-links, "in the sense that the only portions [...] that are possibly related to one another via the log-links consist of roots of unity [...] multiplication by which corresponds, via the log-link, to an 'addition by zero' indeterminacy, i.e., to no indeterminacy!" [H] Thus the possible images lie in the log-shell packet multiplied by the theta values $\underline{\underline{q}}^{j^2}$, and this is our $C$. We take the *largest* reading: everything between an image and $C$ is possible. A smaller set of possible images can only shrink the hull (Proposition 26(a), Proposition 25).

[S] (C4): [IUT3, Rem. 3.9.5(i)] defines the holomorphic hull of a relatively compact $U$ as "the smallest subset of the form" $\lambda\cdot\widetilde{\mathcal{O}}$ that contains $U$, where $\widetilde{\mathcal{O}}$ is the ring of integers of the tensor

packet, which is a direct sum of fields, and each component of $\lambda$ is nonzero. This is the holomorphic integral structure; $\mathcal{O}$ denotes the mono-analytic one.

[S] (C6): the $\Theta_{\mathrm{LGP}}$-link sends the $\Theta$-pilot to the $q$-pilot, and an isomorphism of Frobenioids preserves only isomorphism classes of objects [IUT3, Rem. 3.6.2(i)]; the arithmetic degree is determined up to a positive real multiple [IUT2, Cor. 4.10(v)]. Neither statement is a premise of the proof: the proof cites [IUT3, Rem. 3.6.2(i)] only in Step (xii), where it explains why global realified Frobenioids are needed (Table 2 counts it among the comments), and the statement of [IUT3, Cor. 3.12] and Step (x) of the proof cite [IUT2, Cor. 4.10] only for part (i). [H] We keep $s$ free and use the two statements only to justify this; by Proposition 26(c), fixing $s = 1$ would give the conclusion whenever $\theta$ has nonzero measure. [S] That a positive rescaling of one side alone cannot be controlled by comparisons that are invariant under it is [Pajo, Cor. 9.2]. [H] Here the freedom has a source, namely the two statements just cited.

(C7) is the degree-level content of (IPL): the output is related to the input by a full poly-isomorphism, which on degrees gives a positive multiple and nothing more.

**Definition 4** (readings of (SHE))**.** For a cited theory:

- *weak (SHE)*: there is a map alg from regions to sets of regions with $\mathcal{P} = \mathrm{alg}(C)$ and $y \leq c$ whenever $y \in \mathrm{alg}(c)$. (The possible images are computed from the container, which is common to both sides.)
- *strong (SHE), carrier reading*: $q \in \mathcal{P}$.
- *strong (SHE), value reading*: $\mathrm{vol}(q) \in \mathbb{R}_{\leq \mathrm{vol}(H)}$.
- *(9-1)* [LANA, §§9.2–9.3], in the reading explained below: there is $x \in \mathcal{P}$ of nonzero measure with $\mathrm{vol}(x) = \mathrm{vol}(q)$.

[H] [IUT3, Rem. 3.11.1(iii)] describes (SHE) as the property that the output data and its construction are expressible in terms valid simultaneously for the arithmetic holomorphic structures on both sides. We do not decide between the readings. The value reading is the conclusion rewritten in the language of (xi-d); the carrier reading is the closed loop of [IUT3, Rem. 3.9.5(viii)(ix)] taken literally.

[S] On (9-1): [LANA, §9.2] constructs two isomorphisms of pointed real vector spaces $\eta^q, \eta_S^{\mathrm{anab}} : \mathbb{R}^{\mathrm{val}} \to \mathbb{R}^{\mathrm{ss}}$, where $\mathbb{R}^{\mathrm{ss}}$ consists of the adelic measurable subsets of a large-volume container modulo equality of volume; $\eta_S^{\mathrm{anab}}$ sends the pilot to the class of $\mathrm{LGP} \cdot S$, and $\eta^q$ sends it to the class of $\{q_{\underline{v}}\,\mathcal{O}_{\underline{v}}\}$, where $q_{\underline{v}}$ is the $q$-parameter and $\mathcal{O}_{\underline{v}}$ the ring of integers. The $q$-pilot of [LANA, §4.2(c)] is built from a $2\ell$-th root of $q_{\underline{v}}$, and [LANA, §9.3] reads (9-1) as saying that "at the level of degree (log-volume), the rigidified $q$-pilot is represented in the output regions". [H] We follow this reading: we take the pilot of $\eta^q$ to be the class of the $q$-pilot region, and the regions $\mathrm{LGP} \cdot S$, for the $S$ that [LANA, §9.2] calls suitable, to be the possible images. Two isomorphisms of pointed one-dimensional real vector spaces agree if and only if they agree on the pilot, so, under these identifications, $\eta^q = \eta_S^{\mathrm{anab}}$ for some suitable $S$ if and only if some possible image has the log-volume of the $q$-pilot; this is the fourth item above. Read literally, with $q_{\underline{v}}$ in place of its $2\ell$-th root, the region $q_{\underline{v}}\mathcal{O}_{\underline{v}}$ has, at each bad place, $2\ell$ times the log-volume of the $q$-pilot region there, and (9-1) would compare the possible images with that multiple.

**Proposition 5.** [P] *Weak (SHE) holds in every cited theory: take* $\mathrm{alg}(c)$ *to be the set of possible images* $y$ *with* $y \leq c$*; then* $\mathrm{alg}(C) = \mathcal{P}$ *by (C3).*

So the axioms together with weak (SHE) have the same models as the axioms alone. We keep weak (SHE) in view because it is the reading of (SHE) that the cited statements support: the possible images are computed from the common container. It does not close the gap.

**Proposition 6.** [P] *In a cited theory: the value reading is equivalent to the conclusion; the carrier reading implies (9-1); (9-1) implies the conclusion.*

*Proof.* The first is the definition of $\mathbb{R}_{\le a}$. The second takes $x = q$. For the third, $x \le H \le \mathcal{O}$ by (C4), and on each $P_{v,j}$ the function $A \mapsto w(\log \mu(A) - \log \mu(\mathcal{O}))$ is monotone for $A \subseteq \mathcal{O}$ of nonzero measure; summing gives $\mathrm{vol}(q) = \mathrm{vol}(x) \le \mathrm{vol}(H)$. □

*Remark* 7 (the approximation in (xi-f)). Step (xi-f) allows the construction of $-|\log(\underline{\underline{q}})|$ to hold "perhaps only up to some sort of 'approximation'". Since the set of possible values is a closed half-line, approximate membership from below is the same as membership: $\mathrm{vol}(q) \le \mathrm{vol}(H) + \varepsilon$ for every $\varepsilon > 0$ if and only if $\mathrm{vol}(q) \le \mathrm{vol}(H)$. This is [Higuchi, Lemma 4.1]. [P] Consequently the approximate form fails in every model below in which the conclusion fails.

# 4 A model at one place

[S] The initial Θ-datum of [First, Thm. 1] has $\ell = 7$, so $\ell^* = 3$. Above the split prime 211 it has one selected bad place and one good place; at the bad place the ramification index is $e_{\mathrm{b}} = 105$ and $\mathrm{ord}_{211}(\underline{\underline{q}}_{\mathrm{b}}) = 29/7$. The model of this section keeps $\ell$ and the valuation, and replaces the completion at the bad place by $\mathbb{Q}_{211}$; §5 enlarges the container at that place by a fixed factor, and §6 builds the model from the actual local objects at a finite extension of $\mathbb{Q}_p$ that carries a theta datum. [S] For the datum itself, with all its bad places, the weights of the procession normalization and its container, the two sides of Step (xi) in the reading of [IUT4] are computed in [TC, §8.4]: $-21.80$ against 135.26, so the computed comparison holds there. The failures below concern models in which the container at the bad place is the image itself or a bounded enlargement of it, and the family in $N$.

**Definition 8.** Let $P_j = \mathbb{Q}_{211}^{j+1}$ with the product Haar measure normalized by $\mu(\mathbb{Z}_{211}^{j+1}) = 1$, let $\mathcal{O}_j = \mathbb{Z}_{211}^{j+1}$, $w_j = 1/(7(j+1))$ and $a_j = 0$. Let the automorphisms be the coordinate permutations and the coordinatewise multiplications by $\mathbb{Z}_{211}^{\times}$. Let $B(j,k) = (211^k \mathbb{Z}_{211})^{j+1}$ and $\Lambda = \{B(j,k) : k \ge 0\}$. Put

$$\theta_j = C_j = H_j = B(j, 29j^2), \qquad q_j = B(j, 29), \qquad \mathcal{P} = \{\theta\}, \qquad s = 3/14.$$

The weight $1/(j+1)$ is the packet normalization; the constant factor $1/7$ makes $\mathrm{vol}(q) = -\mathrm{ord}_{211}(\underline{\underline{q}}) \log 211$. No statement below changes if all weights are multiplied by the same positive constant.

Here $\mathcal{O}_j$ is also the ring of integers of $P_j$. The class $\Lambda$ consists of the sets $\lambda \cdot \mathcal{O}_j$ for which the components of $\lambda$ all have the same nonnegative valuation; it is smaller than the class of [IUT3, Rem. 3.9.5(i)]. This does not matter here: $C = \theta$ is itself a possible image and lies in $\Lambda$, so any region of the larger class containing the possible images contains $C$, and the smallest one is $C$ in either class. (The model of §6 uses the full class.)

The log-shell of a nonarchimedean local field $k$ with ring of integers $\mathcal{O}_k$ is $I_k = (p^*)^{-1} \log_k(\mathcal{O}_k^{\times})$, with $p^* = p$ for odd $p$ and $p^* = p^2$ for $p = 2$ [IUT3, Rem. 1.2.2(i)]. For $k = \mathbb{Q}_{211}$ this is $211^{-1} \cdot 211\, \mathbb{Z}_{211} = \mathbb{Z}_{211}$, so (Ind3) does not enlarge anything on this carrier and $C = \underline{\underline{q}}^{j^2} \cdot \mathcal{O}$ is the

possible image itself. For the actual completion $K$ at the bad place, with $e = 105 < 210 = p - 1$, the logarithm maps $1 + \mathfrak{m}_K$ onto $\mathfrak{m}_K$ and $K$ contains no nontrivial $p$-power roots of unity, so $I_K = p^{-1}\mathfrak{m}_K \supsetneq \mathcal{O}_K$. The container is then larger than the image of the $\Theta$-pilot, by a factor that does not depend on the Tate parameter. Theorem 15 below covers enlargements by $211^c$; Theorem 22 treats the actual objects.

**Lemma 9.** [P] *Put* $u = -\frac{29}{7}\log 211 < 0$. *At every stage* $j$, $w_j\big(\log\mu(B(j,29m)) - \log\mu(\mathcal{O}_j)\big) = m\,u$. *Hence* $\mathrm{vol}(q) = u$ *and* $\mathrm{vol}(\theta) = \frac{1+4+9}{3}\,u = \frac{14}{3}\,u$.

*Proof.* $\mu(B(j,k)) = 211^{-k(j+1)}$, so $w_j \log\mu(B(j,29m)) = -\frac{29m}{7}\log 211$. □

Note that $14/3 = \ell(\ell+1)/12$ for $\ell = 7$, the familiar factor. Its reciprocal $12/(\ell(\ell+1))$ is the constant $\chi$ compared in [Fifth], where it is identified as the ratio of the stage-one coefficient $1/(2\ell)$ to the mean of $j^2/(2\ell)$.

**Theorem 10.** [P] *The data of Definition 8 form a cited theory in which the conclusion fails:* $\mathrm{vol}(q) = u > \frac{14}{3}u = \mathrm{vol}(H)$. *Hence none of the conclusion, the carrier reading, the value reading, or (9-1) follows from the axioms of a cited theory, even together with weak (SHE).*

*Proof.* (C1), (C4), (C5) are immediate: $\theta$ is the only possible image, it is a box $B(j,29j^2) \in \Lambda$, so it is the smallest region of $\Lambda$ containing the possible images, and $\mathrm{vol}(q) = u < 0$. (C2): permutations and multiplication by units preserve $B(j,k)$. (C3): $x \le y \le C = \theta$ forces $y = \theta$. (C6): $u = \frac{3}{14}\cdot\frac{14}{3}u$. (C7) with $c = 3/14$. The conclusion $u \le \frac{14}{3}u$ is false because $u < 0$; the readings and (9-1) fail by Proposition 6. □

**Corollary 11** (where the gap lies)**.** [P] *Weak (SHE) holds in every cited theory; in every cited theory the value reading is equivalent to the conclusion; and neither the carrier reading nor the value reading holds in every cited theory.*

*Proof.* Propositions 5 and 6, and Theorem 10. □

[H] So the whole distance between the axioms and the conclusion lies between the weak and the strong readings of (SHE). This is partly a matter of definition, since the value reading is the conclusion rewritten; the content is that the weak reading follows from the axioms and the strong readings do not. In the source, the passage between the two readings is the passage from (xi-e), which invokes (SHE), to (xi-f), which draws the membership. [S] Higuchi locates the same passage for his qualitative package [Higuchi]. In the one-number model of [SL, §5], (SHE) read as a property of the shared data holds both where the inequality holds and where it fails [SL, Theorem 5.7], while (SHE) read as containing the $q$-pilot asserts the 2-cell and is equivalent to the height bound [SL, Remark 5.8, Theorem 5.3]; there membership implies the lax 2-cell and not conversely [SL, Proposition 6.1]. [H] The corollary shows that the log-volume content of the cited results does not close it.

*Remark* 12. The scale $s = 3/14 = 12/(\ell(\ell+1))$ is not chosen: (C6) forces it from the two volumes, which are determined by the valuation of the datum and the Haar measure. The gap $\mathrm{vol}(q) - \mathrm{vol}(H) = \frac{11}{3}|u|$ is what (Ind3) would have to cover. In the notation of [RF, SL], with $k = 1/s = 14/3$ and $a = Q = |u|$, it is the wall $(k-1)a$ (the term is that of [LANA]), and Theorem 10 is the case $E = 0$ of [SL, Corollary 5.4]. Our $s$ is the $m$ of [SL], the $1/k$ of [II, RF, TC, BS] and the $\chi$ of [Fifth]; [lana] writes $s$ for $\ell(\ell+1)/12$. That the slack must come from the hull and not from the orbits of (Ind1) and (Ind2) is also proved in [Takkun] (`hull_exceeds_orbit`, `slack_lower_bound`).

*Remark* 13. (C6) and (C7) can be satisfied, by choosing $s$ and the constants $c$, whenever the volumes involved are negative: they fix a positive ratio and nothing more. This is the content of (IPL) at the level of degrees, and it is why the degree scale is a parameter of the model rather than a constraint on it.

## 5 Any number of places

A cited theory over several places is defined in the same way, with regions indexed by places and vol summed over them; $\ell^*$ is common to all places. Each place carries its own packet system, so the possible images are required to be closed under (Ind1) and (Ind2) chosen independently at the different places. For the label permutations of (Ind1) this is the stronger requirement; the automorphisms of (Ind2) are chosen place by place in either case. An *extra place* is any packet system with $\ell^* = 3$ together with a region $q^{\mathrm{ex}} \leq \mathcal{O}$ of nonzero measure. At an extra place we take the $\Theta$-side image, the container, the hull and the only possible image to be the whole integral structure $\mathcal{O}$, and $\Lambda = \{\mathcal{O}\}$; since $\mathcal{O}$ is then the only possible image, $H = \mathcal{O}$ is also the smallest member of any class that contains $\mathcal{O}$. This is the largest enlargement that (Ind3) permits, and it makes the hull as large as possible.

At the place over 211 we multiply the valuations of both pilots by $N$: $\theta_j^{(N)} = B(j, 29Nj^2)$ and $q_j^{(N)} = B(j, 29N)$. This is what replacing the Tate parameter by its $N$-th power would do; it is a family of data, not a remark of the source. [S] The same family is used in [RF, Theorem 3.6], and on six explicit data the comparison computed in [TC, §10] fails from $N = 3$ (from $N = 4$ for one datum). ([IUT3, Rem. 3.11.1(ii)] replaces only the $q$-pilot by the $q^\lambda$-pilot, which makes the conclusion weaker, not stronger.) The data of this family are not claimed to come from elliptic curves: for actual curves the contributions of the other places need not stay fixed as the valuation of the Tate parameter grows.

**Theorem 14.** [P] *For every finite family of extra places there is $N$ such that the resulting data form a cited theory over all places in which the conclusion fails. In particular this holds with two extra places: one over $\mathbb{Q}_5$ (packets $\mathbb{Q}_5$ at every stage, $\mathcal{O} = \mathbb{Z}_5$, $q$-component $5\mathbb{Z}_5$), and one archimedean place (packets $\mathbb{R}^{j+1}$ with Lebesgue measure, $\mathcal{O}$ the closed unit ball, weights $1/(j+1)$, $q$-component the closed ball of radius $1/2$).*

*Sketch.* Let $A = \sum \mathrm{vol}(\mathcal{O}^{\mathrm{ex}})$ and $B = \sum \mathrm{vol}(q^{\mathrm{ex}}) \leq A$ over the extra places. The total volumes are $N\frac{14}{3}u + A$ and $Nu + B$. The conclusion would require $\frac{11}{3}N|u| \leq A - B$, which fails for large $N$. The axioms are checked place by place; the only new point is that the automorphisms preserve $\mathcal{O}$. □

We now separate the image of the $\Theta$-pilot from the container at the bad place: for a fixed integer $c \geq 0$ let

$$\theta_j^{(N)} = B(j, 29Nj^2), \qquad C_j^{(N,c)} = B\big(j, \max(29Nj^2 - c, 0)\big),$$

(for $c \leq 29N$ the maximum is $29Nj^2 - c$ at the stages $j \geq 1$, and 0 at the stage $j = 0$, which does not enter vol) and let $\mathcal{P}$ consist of the regions $x$ with $\theta^{(N)} \leq x \leq C^{(N,c)}$ at the bad place and $x = \mathcal{O}$ at the extra places, so that the possible images form the whole interval between the image and the container, and the hull is $C^{(N,c)}$. The container exceeds the image in normalized log-volume by $\frac{c}{7}\log 211$, independently of $N$.

**Theorem 15.** [P] *For every integer $c \geq 0$ and every finite family of extra places there is $N$ with $c \leq 29N$ such that these data form a cited theory over all places in which the conclusion fails. For $c > 0$ the image of the $\Theta$-pilot and the container are different regions.*

*Sketch.* The total volumes are $N\frac{14}{3}u + \frac{c}{7}\log 211 + A$ for the hull and $Nu + B$ for the $q$-pilot. The conclusion would require $\frac{11}{3}N|u| \leq \frac{c}{7}\log 211 + A - B$, which fails for large $N$. Closure under (Ind1), (Ind2) holds because the automorphisms fix both ends of the interval and preserve inclusions; every possible image contains $\theta^{(N)}$ and so has nonzero measure. □

# 6 The model on the actual local objects

The models above replace the completion at the bad place by $\mathbb{Q}_{211}$ and let the indeterminacies act by coordinate permutations and units. We now build the model from the actual local objects at a finite extension of $\mathbb{Q}_p$, ramified or not, that carries a theta datum (Definition 19).

## 6.1 Packets and the integral structure

Let $p$ be a prime and $L$ a finite extension of $\mathbb{Q}_p$, with ring of integers $R$, ramification index $e$ and a uniformizer $\pi$. For $j \in \{0,1,2,3\}$ let $I = \{0,\dots,j\}$, let $P_j = L^{\otimes I}$ be the tensor product over $\mathbb{Q}_p$ of $j+1$ copies of $L$, and let $(R_I)^{\sim} \subseteq P_j$ be the integral closure of $\mathbb{Z}_p$ (the ring of integers $\widetilde{\mathcal{O}}$ of the packet). Let $\log_p(R^{\times}) \subseteq L$ be the image under $\log_p$ of the principal units $1 + \mathfrak{m}_R$; it is also the image of $R^{\times}$, since $R^{\times}$ is the product of $1+\mathfrak{m}_R$ and the roots of unity of order prime to $p$, on which $\log_p$ vanishes (a standard fact, not formalized). Let $\log_p(R_I^{\times}) \subseteq P_j$ be the $\mathbb{Z}_p$-module generated by the pure tensors of elements of $\log_p(R^{\times})$ [IUT4, Prop. 1.2]; on each factor $\log_p(R^{\times}) = p^{*} \cdot I_L$, where $I_L$ is the log-shell of §4. The constants $\mathfrak{d}_I$, $a_I$, $b_I$ are those of [IUT4, Prop. 1.1, 1.2], computed from the actual different and ramification of $L$. For $x \in L$ we write $x_{(1)} = x \otimes 1 \otimes \cdots \otimes 1 \in P_j$, and for $\lambda = k/e$ we write $p^{\lambda} \cdot (R_I)^{\sim}$ for $(\pi^k)_{(1)} \cdot (R_I)^{\sim}$.

**Lemma 16.** [P] *$\log_p(R_I^{\times})$ is a compact open subgroup of $P_j$. For every $\mathbb{Q}_p$-linear automorphism $\phi$ of $P_j$ with $\phi(\log_p(R_I^{\times})) = \log_p(R_I^{\times})$ and every $\lambda \in e^{-1}\mathbb{Z}$,*

$$\phi\big(p^{\lambda}(R_I)^{\sim}\big) \subseteq p^{\lfloor \lambda - \mathfrak{d}_I - a_I \rfloor} \log_p(R_I^{\times}) \subseteq p^{\lfloor \lambda - \mathfrak{d}_I - a_I \rfloor} \cdot \pi_I^{-b} \cdot (R_I)^{\sim},$$

*where $\pi_I^{-b}$ is a pure tensor of powers of $\pi$ of total order $-b_I$ [IUT4, Prop. 1.2(ii)].*

We take the packet system with one place, $\ell^{*} = 3$, $P_j$ as above, $\mathcal{O}_j = \log_p(R_I^{\times})$, the Haar measure $\mu_j$ with $\mu_j((R_I)^{\sim}) = 1$, the weight $w_j = 1/[P_j : \mathbb{Q}_p]$ and the adjustment $a_j = w_j \log \mu_j(\mathcal{O}_j)$. [S] By [IUT3, Rem. 3.9.1(i)] the mono-analytic integral structure is the tensor product of the log-shells $I_L$, which differs from $\mathcal{O}_j$ by the scalar $(p^{*})^{j+1}$; and [IUT4, Prop. 1.4(i)] normalizes log-volumes on such tensor products so that $\mu^{\log}((R_I)^{\sim}) = 0$ and $\mu^{\log}(p \cdot (R_I)^{\sim}) = -\log p$.

**Lemma 17.** [P] *For every region $A \subseteq P_j$, $w_j\big(\log \mu_j(A) - \log \mu_j(\mathcal{O}_j)\big) + a_j = w_j \log \mu_j(A)$. Hence the packet log-volume is the one normalized as in [IUT4, Prop. 1.4(i)].*

The identity is algebraic, and holds for any compact open choice of $\mathcal{O}_j$ with $a_j$ defined in the same way; in particular it holds for the tensor product of the log-shells itself.

The indeterminacies act as follows. A permutation $\sigma$ of the labels $I$ permutes the tensor factors. An automorphism of prime-strips at a label, and a copy of Ism at a label, act on the corresponding

factor through a $\mathbb{Z}_p$-linear automorphism $g$ of the log-shell $I_L$, extended $\mathbb{Q}_p$-linearly to $L$; the copies at different labels are independent. We let $g$ run over *all* $\mathbb{Z}_p$-linear automorphisms of $I_L$. [S] [IUT3, Thm. 3.11(i)] describes (Ind2) as induced "by the action of independent copies of Ism [...] on each of the direct summands of the $j+1$ factors appearing in the tensor product". By [IUT3, Prop. 1.2(vi)] the mono-analytic log-shell is constructed by a functorial algorithm from the $\mathcal{D}^{\vdash}$-prime-strip, and the Ism-orbit of isomorphisms constructed there is compatible with the log-shells and their log-volumes; so the indeterminacies of (Ind1) and (Ind2) act on each factor through automorphisms of the log-shell. [H] We do not construct the maps from the groups of the source to $\mathrm{Aut}_{\mathbb{Z}_p}(I_L)$; closure of the possible images under the full group implies closure under their images. [S] For (Ind2) the image is known: by [Fourth, Proposition 9], Ism is $\mathbb{Z}_p^{\times}$ at a finite extension of $\mathbb{Q}_p$ and acts by unit scalars, so the full group is an enlargement; it is the group `tensorAut` of [lana], described there as a deliberate enlargement.

**Lemma 18.** [P] *These actions satisfy the group laws, preserve* $\log_p(R_I^{\times})$*, and hence preserve* $\mu_j$*. The copies of* Ism *in (Ind2), and the automorphisms of prime-strips in (Ind1), act nontrivially for every* $L$*: multiplication by* $-1$ *on the log-shell sends* $1 \in P_0$ *to* $-1$*. If* $L \neq \mathbb{Q}_p$*, the label permutations in (Ind1) act nontrivially as well: for* $\alpha \in L \setminus \mathbb{Q}_p$*, the transposition of the two labels of* $P_1$ *sends* $1 \otimes \alpha$ *to* $\alpha \otimes 1 \neq 1 \otimes \alpha$*; the* $\mathbb{Q}_p$*-linear map* $x \otimes y \mapsto \mathrm{Tr}_{L/\mathbb{Q}_p}(x)\, y$ *separates the two.*

## 6.2 Theta data and regions

**Definition 19** (theta datum)**.** A *theta datum* over $L$ consists of $\tau, i \in L^{\times}$ with $i^2 = -1$ and $|\tau| < 1$, together with $\xi_1, \xi_2, \xi_3 \in L^{\times}$ and $r \in L^{\times}$ such that $\xi_j^7$ is one of $\vartheta(i)/\vartheta(\pm\tau^j i)$, where $\vartheta$ is the theta series of [EtTh, Prop. 1.4] with parameter $\tau$, and $r^{14} = \tau^2$. Put $\mathfrak{q} = \tau^2$ (the Tate parameter).

[P] Since $L$ is complete and nonarchimedean, the series converges and $\vartheta(i) \neq 0$. Moreover $\xi_j^{14} = \mathfrak{q}^{j^2}$, so $|\xi_j| = |\mathfrak{q}|^{j^2/14}$ and $|r| = |\mathfrak{q}|^{1/14}$; for every $N \geq 1$, $(\tau^N, i, \xi^N, r^N)$ is again a theta datum, with Tate parameter $\mathfrak{q}^N$.

**Definition 20** (the model at $L$)**.** Fix a theta datum and $N \geq 1$. For $j = 1, 2, 3$ let $k_j \in \mathbb{Z}$ be given by $|\xi_j|^N = |\pi|^{k_j}$, and put $c_j = \lceil b_I \rceil + \lceil \mathfrak{d}_I + a_I \rceil$ and $m_j = \lfloor k_j/e - \mathfrak{d}_I - a_I \rfloor - c_j$, with $I = \{0, \ldots, j\}$; at the stage $j = 0$, which does not enter vol, use $\xi_1$ in place of $\xi_0$.

- $\theta_j = (\xi_j^N)_{(1)} \cdot (R_I)^{\sim}$ and $q_j = (r^N)_{(1)} \cdot (R_I)^{\sim}$.
- The *horizontal images* at stage $j$ are the sets $\phi\big((u\,\eta_j^N)_{(1)} \cdot (R_I)^{\sim}\big)$, where $\eta$ runs over the theta data $(\eta_1, \eta_2, \eta_3)$ with the same $\tau, i$, $u$ over the elements of $L$ with $|u| = 1$, and $\phi$ over the $\mathbb{Q}_p$-linear automorphisms of $P_j$ preserving $\log_p(R_I^{\times})$. The *vertical images* are the sets $(u\,\eta_j^N)_{(1)} \cdot \mathcal{I}_j$, where $\mathcal{I}_j$ is the additive subgroup of $P_j$ generated by the pure tensors of elements of $I_L$.
- $C_j = p^{m_j} \cdot \log_p(R_I^{\times})$.
- $H_j$ is the smallest set of the form $\lambda \cdot (R_I)^{\sim}$, $\lambda \in P_j^{\times}$, containing $C_j$.
- $\mathcal{P}$ consists of the regions $x$ with $b \leq x \leq C$ for some $b$ whose components are horizontal images; $\Lambda$ is the class of all sets $\lambda \cdot (R_I)^{\sim}$, $\lambda \in P_j^{\times}$, as in [IUT3, Rem. 3.9.5(i)]; and $s = 3/14$.

The container is a $p$-power multiple of the log-shell packet, as the source requires (Remark 3). Lemma 16, applied to the automorphisms $\phi$, places the horizontal images in the middle term

$p^{\lfloor k_j/e-\mathfrak{d}_I-a_I\rfloor}\log_p(R_I^\times)$ of [IUT4, Prop. 1.4(iii)]. For the vertical images we only use their containment in the right-hand term, and passing from that term back to a multiple of the log-shell packet costs the factor $p^{-c_j}$. A larger container gives a larger hull and so works against the failure of the conclusion; it fails nonetheless.

**Lemma 21.** [P]

*(a)* *$C_j$ contains all horizontal and vertical images, and every $\mathbb{Q}_p$-linear automorphism of $P_j$ preserving $\log_p(R_I^\times)$ maps $C_j$ onto itself.*

*(b)* *$H_j$ exists and is of the form $\lambda\cdot(R_I)^\sim$; it is contained in $p^{m_j}\pi_I^{-b}\cdot(R_I)^\sim$.*

*(c)* *$\mathcal{P}$ is closed under the actions of Lemma 18.*

*(d)* *$w_j\log\mu_j(H_j)\le(b_I+\mathfrak{d}_I+a_I+1+c_j)\log p+N\log|\xi_j|$.*

**Theorem 22.** [P] *Let $L$ be a finite extension of $\mathbb{Q}_p$ and fix a theta datum over $L$.*

*(a)* *$\mathrm{vol}(\theta)=\frac{N}{3}\log|\mathfrak{q}|$ and $\mathrm{vol}(q)=\frac{N}{14}\log|\mathfrak{q}|$, and*

$$\mathrm{vol}(H)-\mathrm{vol}(\theta)\le E_L:=\frac{1}{3}\sum_{j=1}^{3}\big(b_{I_j}+\mathfrak{d}_{I_j}+a_{I_j}+1+c_j\big)\log p,$$

*a constant that depends only on $L$ (not on the theta datum, nor on $N$).*

*(b)* *There is $N\ge1$ for which the data of Definition 20 form a cited theory in which the conclusion fails.*

*(c)* *For every finite family of extra places there is $N\ge1$ for which the same holds over all places, with the extra places treated as in §5 and $s$ the ratio of the total volumes of $q$ and $\theta$.*

*(d)* *There are $\mathfrak{q}_\mathrm{b},\rho,i\in\mathbb{C}_{211}$ with $\mathfrak{q}_\mathrm{b}^7=211^{29}$, $\rho^{14}=\mathfrak{q}_\mathrm{b}$ and $i^2=-1$, such that for $L=\mathbb{Q}_{211}(\rho,i)\subseteq\mathbb{C}_{211}$: the ramification index of $L$ is divisible by 98, $(\tau,i,\xi_j,r)=(\rho^7,i,\rho^{j^2},\rho)$ is a theta datum over $L$ with Tate parameter $\mathfrak{q}_\mathrm{b}$, and (b) and (c) hold for it (with the two extra places of Theorem 14 in (c)). Here $L\ne\mathbb{Q}_{211}$, so the label permutations of (Ind1) act nontrivially.*

*Sketch.* By Lemma 17, volumes are computed with $\mu^{\log}((R_I)^\sim)=0$ and $w_j=1/[P_j:\mathbb{Q}_p]$, so a single factor $x_{(1)}$ contributes $\log|x|$ at every stage; this gives the two volumes in (a). For the hull, Lemma 21(d) and $(k_j/e)\log p=-N\log|\xi_j|$ give the bound. The conclusion would require $\frac{N}{14}\log|\mathfrak{q}|\le E_L+\frac{N}{3}\log|\mathfrak{q}|$, i.e. $\frac{11}{42}N|\log|\mathfrak{q}||\le E_L$, which fails for large $N$; large $N$ also gives $C,H,q\le\mathcal{O}$ (these hold once the exponents $\lfloor k/e-\mathfrak{d}_I-a_I\rfloor$ attached to $\xi_j^N$ and to $r^N$ exceed $2|c_j|$). The axioms (C1)–(C7) follow from Lemmas 18 and 21. In (c) the extra places contribute constants, as in Theorem 14. In (d), $|\rho|^{98}=211^{-29}$ and $\gcd(29,98)=1$; in particular $|\rho|$ is not in the value group $211^{\mathbb{Z}}$ of $\mathbb{Q}_{211}$, so $\rho\notin\mathbb{Q}_{211}$. □

*Remark* 23. [H] Which choices are ours. The Tate parameter $\mathfrak{q}_\mathrm{b}$ in (d) has valuation 29/7, which in §4 is the valuation of the element $\underline{q}_{\underline{=}\mathrm{b}}$ of [First]; here it is used as the Tate parameter, so the $q$-pilot $r=\rho$ has valuation 29/98, not 29/7. It is not claimed to be the $q$-parameter of the curve of [First], and $L$ is not claimed to be a completion of its number field. The family in $N$ is a family of theta data, not a family of elliptic curves. The possible images are read maximally (Remark 3). The groups of (Ind1), (Ind2) are replaced by all automorphisms of the log-shells. The first two are choices of data. Each of the others makes the hull larger or the closure requirement stronger, and none of them lowers the hull.

# 7 The level of Step (xi)

[S] [IUT3, Rem. 3.11.1(iii)] describes (IPL) as follows: the output data are linked, via full poly-isomorphisms of $\mathcal{F}^{\Vdash\blacktriangleright\times\mu}$-prime-strips induced by operations in the algorithm, to the $q$-pilot prime-strip; at a more explicit level, the linking isomorphisms of the intermediate prime-strips are composites of the inverses of the first two lower horizontal arrows of the diagram of [IUT3, Rem. 3.10.2], its first vertical arrow and its three upper horizontal arrows. About (SHE) it says: "In passing, we observe that this property 'SHE' may be understood, in a slightly more concrete way, as corresponding to the fact that the chain of (sub)quotients considered in Remark 3.9.5, (viii), (ix), forms a closed loop." [IUT3, Rem. 3.9.5(ix)] states that the chain that begins at the $q$-pilot prime-strip, passes through the $\Theta$-link, through the poly-isomorphisms induced by (sQ1), (sQ2), and finally through (sQ3), (sQ4), forms a closed loop up to formal quotient indeterminacies, before (sQ5) is applied.

**Theorem 24.** [P] *On every cited theory, over one place or several, one can place the following structure: a set of possible images generated, as in [IUT3, Rem. 3.11.1(iii)], from the image of the $\Theta$-pilot by the steps (Ind1), (Ind2), (Ind3) and by passage to a hull (for which the container, the largest choice, is used), and contained in $\mathcal{P}$ (equal to $\mathcal{P}$ when every possible image contains $\theta$); a link of each of them to the $q$-pilot prime-strip, by the inverse of the composite of the six arrows of (IPL) followed by the $\Theta$-link; and a chain through (sQ1)–(sQ4) that closes before (sQ5) with trivial holonomy. If some possible image in this structure has nonzero measure and the volume of the $q$-pilot, the conclusion holds.*

[H] In this theorem the prime-strips are modelled by a groupoid that records only the sizes of their components; the log-volume is computed from the regions alone. We route the link to the $q$-pilot prime-strip through the $\Theta$-link, as the loop of [IUT3, Rem. 3.9.5(ix)] does. The loop closes with trivial holonomy by construction: the arrow (sQ4) is taken to be the inverse of the composite of the other arrows (§11). The theorem shows that the structure the source names can be placed on every model, not that it constrains the models. [S] [BS, §6.2] reads the same passage of the source as closing the loop at the level of structures. [H] The holonomy here is that of this chain of prime-strips; it is not the monodromy of [SL, Definition 3.6], which at the level of degrees is the scale $s \neq 1$ of Proposition 26(c). Its point is the following. In all the models of §§4–6 the loop closes, the (IPL) composite exists, and weak (SHE) holds; but the conclusion fails, so by Proposition 6 no possible image of nonzero measure has the volume of the $q$-pilot. The closed loop to which [IUT3, Rem. 3.11.1(iii)] relates (SHE), together with (IPL), does not yield the membership required in (xi-f).

[S] The source itself separates the formal properties from what the estimate needs. [IUT3, Rem. 3.11.1(vi)–(viii)] considers naive versions $q \mapsto q^{\lambda}$ of the $\Theta$-link and observes that the "trivial multiradial representation" $\{q^{\lambda^n}\}_{n\in\mathbb{Z}}$ "does indeed satisfy the formal analogues of the IPL and SHE properties" but "is useless from the point of view of applications such as the log-volume estimates given in Corollary 3.12": it allows indeterminacies in the value group portion and holds the unit group portion rigid, whereas the estimates rely on the value group portions being held rigid, on the unit group portions being subject to (Ind1), (Ind2), (Ind3), and on the use, via the log-Kummer correspondence, of the intertwining of the additive and multiplicative structures. [H] The models are not of the trivial kind: the images of the two pilots are fixed regions with fixed volumes, and the possible images are subject to (Ind1), (Ind2), (Ind3) on the log-shells. At the level of log-volumes the log-Kummer correspondence enters the cited results through the (Ind3) bound of [IUT3, Thm. 3.11(ii)], which is (C3), and through the compatibility of log-volumes with the log-links stated at the end of [IUT3, Thm. 3.11(ii)] (cf. [IUT3, Prop. 3.9(iv)]), which does not

move the container (§9). So the features that the source names as separating its algorithm from the trivial one are present in the models at this level, and the conclusion still fails.

[S] The proof of [IUT3, Cor. 3.12] (p. 175 of the consulted version) observes that "we may restrict our attention to possible images of a Θ-pilot object that correspond to data [i.e., collections of regions] that may be interpreted as an $\mathcal{F}^{\mathbb{H}\blacktriangleright}$-prime-strip."

**Proposition 25.** [P] *In a cited theory, let $\mathcal{P}' \subseteq \mathcal{P}$ be any subfamily, and let $H'$ be a smallest region of the class $\Lambda$ containing every member of $\mathcal{P}'$. If $H'$ has nonzero measure, then* $\mathrm{vol}(H') \leq \mathrm{vol}(H)$*; so if the conclusion fails, then* $\mathrm{vol}(H') < \mathrm{vol}(q)$.

*Proof.* $H$ is a region of $\Lambda$ containing the members of $\mathcal{P}'$, so $H' \leq H \leq \mathcal{O}$, and vol is monotone on regions of nonzero measure inside $\mathcal{O}$. □

Thus the restriction can only lower the right-hand side of the inequality.

# 8 The two horns

**Proposition 26.** [P] *In a cited theory over any finite set of places:*

*(a) if every $C_{v,j}$ lies in $\Lambda$ and $\theta$ has nonzero measure, then* $\mathrm{vol}(H) \leq \mathrm{vol}(C)$ *for* every *admissible $\mathcal{P}$; hence if* $\mathrm{vol}(C) < \mathrm{vol}(q)$ *the conclusion fails, however the possible images are read;*

*(b) if $\theta = q$, the conclusion holds, and $s = 1$;*

*(c) if $s = 1$ and $\theta$ has nonzero measure, the conclusion holds; hence $s \neq 1$ in every cited theory in which $\theta$ has nonzero measure and the conclusion fails.*

*Proof.* (a) $C$ is a region of $\Lambda$ containing all possible images, so $H \leq C$ by minimality, and vol is monotone. (b) $q = \theta \in \mathcal{P}$ is the carrier reading; and $\mathrm{vol}(q) = s\,\mathrm{vol}(q)$ with $\mathrm{vol}(q) \neq 0$ gives $s = 1$. (c) $\mathrm{vol}(q) = \mathrm{vol}(\theta)$, and $\theta \leq H \leq \mathcal{O}$ by (C1) and (C4), so $\mathrm{vol}(\theta) \leq \mathrm{vol}(H)$ by monotonicity. □

[H] This is the dilemma in its simplest form. Keeping the factor $j^2$ on the Θ-side object, as the source does, the cited results do not give the conclusion. Removing it, by identifying the two pilots, gives the conclusion, but with degree scale 1; the resulting inequality is then between two equal quantities and says nothing about heights. The second case is the identification discussed in [SS, §2.2].

# 9 Items not taken as axioms

[P] Proposition 26(a) shows only this: with the container and the numerical data fixed, no reading of the possible images inside the container closes the gap. A further statement could still make the conclusion derivable in other ways, for instance by forcing $\mathrm{vol}(q) \leq \mathrm{vol}(C)$, by constraining the scale $s$, or by excluding the data of our models. [H] For each cited item not taken as an axiom we record why we believe it does none of these; this is an argument about the source, not a theorem.

- [IUT3, Thm. 3.11(iii)] (compatibility with the $\Theta^{\times\mu}_{\mathrm{LGP}}$-links): the Kummer isomorphisms of [IUT3, Thm. 3.11(ii)], and the constructions that go with them, are equivariant with respect to arbitrary automorphisms of the domain and codomain of these links, up to (Ind1), (Ind2), (Ind3). The statement does not mention log-volumes. At that level it is the closure under (Ind1), (Ind2) and the bound (Ind3), which are (C2) and (C3); it does not move the container, and it gives no reason to fix $s$.

- [IUT3, Prop. 3.9(iv)] and the end of [IUT3, Thm. 3.11(ii)] (compatibility of log-volumes with the log-link): they do not move the container.

- [IUT3, Rem. 3.9.5(vii)] (the hull as the local portion of an arithmetic vector bundle; determinants and tensor powers): by (Ob3-3) there, the step of applying determinants "may be omitted in discussions in which one is only interested in computing log-volumes"; by (Ob4), comparison with $M$-th tensor powers causes no substantive problem, and a tensor power multiplies both sides of the inequality by the same factor.

- Step (xii) of the proof, with [IUT3, Rem. 3.6.2(i)] (the global realified Frobenioids): working only with local Frobenioids would leave an indeterminacy between an open submodule of a log-shell and its $p_v^N$-multiples. In the models the regions are fixed subsets of the packets and the volumes are summed over all places, so no such indeterminacy arises; the step does not move the container, and it gives no reason to fix $s$.

- [IUT3, Rem. 3.9.6] (product formula, "sufficiently small" regions): these concern degrees of global elements; the degree of the $q$-pilot is fixed by the local values of the Tate parameter.

- [IUT3, Rem. 2.4.2(iv)–(vi), Rem. 3.9.7]: they restrict the possible images to those interpretable as prime-strips, which can only shrink the hull (Proposition 25).

- [IUT3, Rem. 3.11.3, Rem. 3.10.1, Rem. 1.2.2, Rem. 1.5.4]: explanations of method and of why Theorem 3.11 holds; they do not change the container.

- [IUT3, Rem. 3.12.2(ii), (iv), (v)] (cited forward in (xi-a), (xi-c), (xi-d)): an interpretive summary of the proof just given—the $\Theta$-link as a gluing isomorphism, the comparison of the 0- and 1-columns, and a toy model of the (xi-e)/(xi-f) inference itself. Its two assertions with log-volume content are that log-link compatibility of log-volumes is the same in the two columns (already [IUT3, Prop. 3.9(iv)] and the end of [IUT3, Thm. 3.11(ii)], item 2 above) and that the $\mathcal{O}^\times$-indeterminacy in relating regions to arithmetic line bundles is absorbed by the holomorphic hull (already [IUT3, Rem. 3.9.5(vii)–(x)]). The remaining point, that the $q$-pilot has no multiradial analogue, is the reason $\mathrm{vol}(q)$ enters Definitions 1–2 as a fixed value while the $\Theta$-side varies in $\mathcal{P}$. No separate log-volume statement remains.

- [IUT3, Rem. 1.5.2(i)–(iii)]: the mono-analytic log-shell of [IUT3, Thm. 1.5(iii)–(v)], which is a premise as cited, viewed as a multiradial container. The remark explains the construction; it asserts no log-volume value beyond Theorem 1.5 and Theorem 3.11, both assumed.

- [IUT3, Rem. 3.8.1]: records that the $\Theta^{\times\mu}_{\mathrm{LGP}}$- and $\Theta^{\times\mu}_{\mathfrak{lgp}}$-links send a $\Theta$-pilot object to a $q$-pilot object. This is the setup of [IUT3, Def. 3.8(ii)] as stated in (C6); the remark contains no assertion about their log-volumes.

- [IUT3, Rem. 3.9.3]: the procession-normalized log-volume must use equal weights, not a weighted average. This is already built into Definition 1(iii), where the procession-normalized volume is $\frac{1}{\ell}\sum_j$ (sources: [IUT3, Rem. 3.1.1(ii)–(iv)], [IUT3, Prop. 3.9(i)(ii)]).

- [IUT3, Rem. 3.9.2]: a mono-analytic algorithm recovering the subquotients of [IUT3, Prop. 3.9(iii)] from the effect of multiplication on log-volumes ([IUT3, Prop. 3.9(ii)]). Recoverability is not a further assertion about the values; the equality of the global arithmetic degree with a log-volume is [IUT3, Prop. 3.9(iii)], already used.

The remaining cited items are those marked background, comment or figure in Appendix A. They are the reasons Theorem 3.11 holds (cyclotomic rigidity, conjugate synchronization, the Kummer theory of mono-theta environments, in Steps (ii), (v)–(viii)), comments after the conclusion (Step (xii), (xi-h)), and the three figures. Theorem 3.11 is assumed at the strength of the source, so no separate log-volume content is taken from them; any that one of them were found to carry would have to be added to Definition 2 explicitly.

## 10 Relation to other work

[S] Scholze and Stix [SS] argue that the passage to Corollary 3.12 requires a consistent identification of copies of $\mathbb{R}$, under which a factor of order $\ell^2$ appears and the inequality becomes empty. [H] Proposition 26(b) is a formal counterpart of that horn. [S] The same manuscript notes that, under its simplifications (such as identifying identical copies of objects along the identity), the critical [IUT3, Thm. 3.11] does not become false but becomes trivial [SS, footnote 12]. [H] In Figure 1, that simplification is the identification of the two pilots, which moves a model to the structures with $s = 1$ (Proposition 26(b)).

[S] Mochizuki [Rpt, §12] agrees that the correspondence $\{\underline{\underline{q}}^{j^2}\} \mapsto \underline{\underline{q}}$ is incompatible with degree computations, and states that "the situation in (AD) occurs whenever one works within a single hol. str./ring theory". [P] In the models the degree scale is not 1 (Proposition 26(c)); at one place it is 3/14. [H] So the models do not identify the degrees of the two pilots: both lie in one packet system, measured by one log-volume, and their degrees are related by the factor $s \neq 1$. [S] In the same section, following (LbLV), "[t]he multiradial algorithms of [IUTchIII], Theorem 3.11, then provide an alternative way to compute the log-volume of the $\Theta$-pilot object in terms of the $q$-hol. str., up to certain indeterminacies"; [H] our $\mathcal{P}$ and $H$ are that computation. [S] In [Rpt, §11] the multiradial algorithms and the inequalities derived from them are said to be "essentially global in nature — i.e., not just a sort of result of summing up essentially local phenomena", and the compatibility of [IUT3, Thm. 3.11] with the correspondence in the $\Theta$-link "cannot be ascribed solely to purely local indeterminacies such as (Ind1, 2, 3)". [H] Our volumes are sums over places, which by [IUT3, Prop. 3.9(iii)] compute global arithmetic degrees, and the degree scale $s$ is a global quantity; so the global degree is present in the models. [S] The proof cites [IUT3, Thm. 3.11(iii)], headed "($\Theta^{\times\mu}_{\mathrm{LGP}}$-Link Compatibility)", which states that the Kummer isomorphisms of [IUT3, Thm. 3.11(ii)], and the constructions that go with them, are compatible with the horizontal arrows, that is, with the $\Theta^{\times\mu}_{\mathrm{LGP}}$-links, up to (Ind1), (Ind2), (Ind3), in the sense that they are equivariant with respect to arbitrary automorphisms of the domain and codomain of these arrows; it does not mention log-volumes. [H] At the level of log-volumes we read it as (C2) and (C3) (§9); it states no relation between the log-volume of the $q$-pilot and those of the possible images. Whether the compatibility of [Rpt, §11] contains more than this at the level of log-volumes, and if so where the source states it, we do not decide.

[S] In his comments on the manuscript of Scholze and Stix, Mochizuki writes that the relationship between the ring structures on the two sides of the $\Theta$-link "is computed by embedding, at the expense of introducing certain indeterminacies, objects of interest (such as various versions of $\Theta$- and $q$-pilot objects) on opposite sides of the $\Theta$-link into a common container", and that "Since this container is a single container, which, in particular, admits a single well-defined log-volume function on a certain collection of its subsets, there is never any issue of 'requiring that a loop' (as in the displayed diagram of §2.2) 'commute' " [Cmt18a, (C17)]. In the later comments [Cmt18b, (C14)] he grants that simple linear relationships exist between the copies of $\mathbb{R}$ arising via the $\Theta$-link, and adds that, when indeterminacies are involved, the relationship between log-volumes of

regions subject to and not subject to them "is much more complicated and depends nontrivially on the geometry of the particular region under consideration". It continues: "In particular, this relationship is highly non-linear", and refers for an elementary example to [Rpt, (LVEx)], where passing from a region of $\mathbb{R}^2$ to its symmetrization under an automorphism of order two changes the log-volume from $\log(4a+2b)$ to $\log(4a+4b)$. [H] Our models have the first two features: the two pilots are regions of the same packets, measured by one Haar log-volume, and their degrees are related linearly, by (C6). As for the third, the excess of the hull over the image of the $\Theta$-pilot is 0 in §4, and in §6 it is at most $E_L$, a constant computed from the different and the ramification of $L$ and independent of $N$ (Theorem 22(a)); in the elementary example it is less than $\log 2$. What the models show is that such a bounded excess yields weak (SHE) and not the membership: the gap between the log-volumes of the $q$-pilot and of the image of the $\Theta$-pilot grows linearly with $N$ (Theorem 22(a)), so the conclusion would need an excess that grows with it. The cited results do not state, at the level of log-volumes, whether the relationship that (C14) describes provides such growth in the source.

[S] Mochizuki's report on formalization [Form, §4] introduces an intermediate statement and notes that "3.11.5 (=3.11+Rmk. 3.9.5)" "never appears in [IUTchI-IV]"; the formalized part takes as an input the arrow (SHE), which takes the input data of the algorithm (HDD) "to be the data given by the $q$-pilot constructed from the etHT". [H] At the level of log-volumes this is the carrier reading of Definition 4: the $q$-pilot is fed to the algorithm that produces the possible images. Our result shows that this input is not a consequence of the cited results at that level, whereas the weak reading is (Corollary 11). [S] The Lean port of D. Richards [Ri] states this "3.11.5" as the proposition that $-|\log q|$ is at most the log-volume of (HDD)$\circ$(SHE), and leaves its proof as "the content of Stage 2" (`lean/IUT/LeanForm.lean`); in `lean/IUT/StageTwo.lean` it derives 3.11.5 from a structure for [IUT3, Thm. 3.11] with a field `theta_link` identifying the region of the $\Theta$-pilot with that of the $q$-pilot. [H] In Figure 1 that field is the identification of Proposition 26(b). [H] In Figure 1, 3.11.5 corresponds to the intersection of the class of models of the cited results with that of strong (SHE).

[S] The interim report of Project LANA [LANA, §10.5] states that the project does not at present have a proof of (9-1). [P] Theorem 10 shows that (9-1), in the reading of §3, does not follow from the log-volume content of the statements cited in the proof of [IUT3, Cor. 3.12].

[S] Joshi [JoshiLANA] states that the proof as published is incomplete and that the missing ingredient is supplied by his arithmetic Teichmüller theory [JoshiIII, JoshiIV]. [H] Our result is consistent with the first statement; we say nothing about the second.

[S] The present note uses only the explicit initial $\Theta$-datum of [First, Thm. 1]; it does not use the source application of Theorems 5 and 6 of [First], which version 0.1.7 of that report withdrew. The local quantities of [IUT4] at such data were computed in [Second, Third], the tuple structure of the tensor packets is described in [Fourth], and the constant $12/(\ell(\ell+1))$, which is the degree scale of our one-place models, is compared with $12/\ell^2$ in [Fifth]. None of these reports proves or refutes the inference from [IUT3, Thm. 3.11] to [IUT3, Cor. 3.12] studied here.

[S] Higuchi [Higuchi] formalizes the qualitative package of Step (xi): a set of abstract labels, a common container $C$ with a label map, an input point $x_q \in C$, a set $O_\Theta \subseteq C$ of output possibilities, a hull operator $H$ (extensive, monotone, idempotent, otherwise arbitrary), an arbitrary map $r\colon C \to \mathbb{R}$, and the condition $r(H(O_\Theta)) = \mathbb{R}_{\le u}$ [Higuchi, Def. 3.1]. The countermodel is $C = \{*\} \times \mathbb{R}$ with $r$ the projection, $x_q = (*, 0)$, $O_\Theta = \{t \le -1\}$ and $H$ the identity [Higuchi, Thm. 5.1]. The paper also shows that the approximation in (xi-f) does not weaken the inclusion (Remark 7), and that the repaired bridge must be equivalent to the existence of a point of $H(O_\Theta)$ with value $r(x_q)$ [Higuchi, Prop. 6.1], which is the abstract form of (9-1). In that abstraction the log-volume is a function on

the points of the container rather than a measure of regions, the hull is not required to be minimal in any class of regions, the input value is not related to the output by any degree scale, and it is not required to be negative. [H] Our axioms impose all of these, as the source does, and the conclusion still fails.

[S] Pajo [Pajo] works with two typed log-volume domains, embeddings $\iota_q, \iota_\Theta$ into a common container, a comparison certificate $\iota_q(\ell_q) \preceq_C \iota_\Theta(\ell_\Theta)$ and numerical evaluations $F_q, F_\Theta$; the countermodel rescales $\iota_\Theta$ by $1/4$, so that the comparison in the container holds while $F_q(\ell_q) \le F_\Theta(\ell_\Theta)$ fails [Pajo, Thm. 6.1, §7], and a parametric version rescales by any $\lambda \le a/b$ [Pajo, Thm. 9.1]. Among the additional data that defeat this countermodel it lists a common numerical chart [Pajo, §11.1]. [H] In our setting such a chart is part of the source: both pilots and the hull are measured by the same log-volume on the same packets [IUT3, Prop. 3.9]. What fails in our models is therefore not the passage from a container comparison to a number, but the container comparison itself: the image of the $q$-pilot is not among the possible images, and its volume exceeds that of the hull.

[S] The file `Iut/WallEquiv/NonDerivable.lean` of the public repository lana-agents/iut of Project LANA [lana] collects what the repository proves at the level of [IUT3, Thm. 3.11] into the package `Thm311Package`. It proves in Lean that such packages arise from genuine initial $\Theta$-data (`ofThetaData`), that Step (xi) is equivalent to $(s^{-1} - 1)\, a \le V$ in our notation (`xi_iff`), and that for all numbers compatible with the fields, for instance $\ell = 7$, $a = 1$, $V = 1$, the fields alone do not give Step (xi); it states that the fields exhaust what the repository proves as an audited claim, not a theorem. [H] The gap of Corollary 11 sits where its sharp form `xi_iff` sits; the differences lie in how the axioms are chosen and how the models are built (Table 1). The two readings of (SHE) are separated earlier in `Iut/Wall/Constraints.lean` of the same repository (§1). The results of this note were obtained independently of these files, and they were made public first.

[S] Zeng [Zeng] (October 2026) reads the holomorphic hull of [IUT3, Rem. 3.9.5] summand by summand and studies the index that this introduces; the hull of our models is the smallest multiple of the integral structure that contains the container ((C4)). Geiger [Geiger] (August 2026) records that the decisive membership of (xi-f) is stated to follow formally from a pair of declared properties rather than derived; Corollary 11 locates the same membership at the level of log-volumes.

[S] Among our published notes, [RF] treats 36 readings of Corollary 3.12 and the conservation of the height load, [SL] views Step (xi) as a lax 2-cell in homotopy type theory, [II] gives the same comparison in elementary notation, [BS] separates a bridge from its size, and [TC] reads the non-derivability of lana-agents/iut as a statement about the specification of Theorem 3.11 and places the key in an implementation of the multiradial algorithm. The present note starts instead from the results that the proof cites.

[H] The results of Higuchi and Pajo are complementary to ours. Theirs show that an abstraction without numerical structure cannot yield the inequality; ours shows that adding the numerical structure stated in the cited results does not help, and locates what is missing at the membership of the $q$-pilot among the possible images, that is, at (xi-f).

## 11 Machine verification

The Lean sources are in the folder `where-311-sits/lean` at tag v0.28.0 of `https://github.com/turahigashi/iut-explicit-initial-theta` and are archived at 10.5281/zenodo.24044582, together with `Verify.lean`, which checks the axioms of every declaration in the modules that carry the results, and `scripts/mutations.py`, which reproduces the mutations described below. All statements marked [P] are proved in Lean 4 (v4.30) with Mathlib. The packet system of Definition 1 carries the Haar measure of Mathlib; the 211-adic and 5-adic Haar measures are Mathlib's on

$\mathbb{Q}_p$, and in §6 the Haar measure is the one normalized on the integral closure of the tensor packet. The tensor packets, the constants of [IUT4, Prop. 1.1, 1.2], the extension of log-shell automorphisms to the field, the theta series of [EtTh, Prop. 1.4] and its analytic properties over complete nonarchimedean fields are constructed, not assumed. Two groups of modules are named after the interim report of Project LANA [LANA]: `independence/Cor312LanaTarget.lean` states its equality (9-1) in the cited theory (Proposition 6), and the twelve modules `relative/LanaStep6*.lean` are named after Step 6 of [LANA, §6.2(g)]. The latter formalize the generic Kummer construction of [LANA, §6.1] and the Kummer-log realization of that step; they are used by some of the modules in `theorem311/` and by the further models in `independence/Cor312SourceBound*.lean`, and not by the statements of this paper. All of these modules were written by the author and contain no code from the repositories of Project LANA. Every declaration depends only on `propext`, `Classical.choice` and `Quot.sound`; no `sorry`, no `native_decide`, and no new axiom is used. The equivalence of Remark 7 and the non-derivability of the approximate form are proved as well. In §6 the logarithm of the units is taken on the principal units (§6.1), and the theta values are values of the theta series of [EtTh, Prop. 1.4] at the chosen parameters; they are not identified in Lean with the theta values reconstructed by the algorithms of the source.

Each mutation below breaks the declaration it targets. The script lists the edits; where an unchanged proof uses the old value for a reason unrelated to the property under test, the edit adapts that proof, so that only the property under test can fail. For the models of §§4–5: widening the hull to the whole integral box breaks exactly the minimality in (C4); adding $q$ to the possible images breaks exactly the bound $x \le C$ in (C3) and the containment $x \le H$ in (C4); removing $j^2$ from the $\Theta$-side image (and correcting its volume) breaks exactly the numerical inequality in the proof of Theorem 14; dropping the nonzero-measure hypothesis breaks exactly the step of Proposition 6 that uses monotonicity; for Theorem 15, dropping the lower end of the interval of possible images breaks exactly the statement that the possible images have nonzero measure, on which the proof of (C7) rests; the closure under (Ind1), (Ind2) and the minimality of the hull survive, since the container is invariant under them and is the hull. For §6: tightening the exponent of the container by one breaks the inclusion of the outer bound in the container; doubling the adjustment breaks Lemma 17; dropping $c_j$ from the bound of Lemma 21(d) breaks that bound; replacing 3/14 by 3/13 breaks (C6); replacing $-1$ by the identity breaks the nontriviality in Lemma 18, and so do replacing $1 \otimes \alpha$ by $\alpha \otimes \alpha$ and dropping the hypothesis $L \neq \mathbb{Q}_p$; stating the monotonicity of $p^m \cdot \log_p(R_I^{\times})$ in $m$ in the opposite direction makes that statement fail. For Propositions 5 and 26(c): computing the possible images from the image of the $\Theta$-pilot instead of the container breaks the first, and dropping the nonzero-measure hypothesis breaks the second. For §7: choosing the arrow (sQ4) that does not move components breaks the triviality of the holonomy, so it is a property of the chosen arrow and not automatic; removing the container bound from the (Ind3) step breaks the inclusion of the generated possible images in $\mathcal{P}$.

## Changes in version 1.2.0

This version corrects the citation inventory; no statement proved here and no Lean source changes. An independent recount (10 November 2026) found that the numbers "152 citations of 61 targets" in versions 1.0.0–1.1.1 came from an inventory extracted by pattern matching from a Japanese translation of [IUT3]: it grouped [IUT1, §§4, 5, 6] as one target and missed twelve textual occurrences in the English original, among them eight mentions of Theorem 3.11. Section 2 now fixes the counting rule and reports 164 occurrences of 63 targets; split by the old grouping, the target set is the same 61, and no role assignment changes. Table 2 now names all fifteen premise targets,

including the twelve remarks previously left unnamed (Rems. 1.2.2, 1.5.2, 1.5.4, 2.4.2, 3.8.1, 3.9.2, 3.9.3, 3.9.6, 3.9.7, 3.10.1, 3.11.3, 3.12.2), and Appendix A lists every target with its count and role, as the abstract has claimed since version 1.0.0. Section 9 now disposes of the five premise remarks it previously did not mention (Rems. 1.5.2, 3.8.1, 3.9.2, 3.9.3, 3.12.2). The three sources not cited in Corollary 3.12 directly ([IUT3, Prop. 3.5(ii)], [IUT3, Prop. 3.2(ii)], [IUT3, Rem. 3.9.1]) are now marked † in §2; only the first was disclosed before. The consulted author versions in the bibliography are corrected to December 2020 for [IUT2] and April 2020 for [IUT4], as printed on the copies used. The Lean sources are unchanged from tag `v0.25.1`; the recorded runs of §11 (7 November 2026) therefore stand, and this version's artifact is tagged `v0.28.0`. Prior work: the omission of [LANA] in 1.0.0 and of [Higuchi] and [Pajo] in versions before 1.1.0, and the misdated comparison with [LANA] corrected in 1.1.1, are recorded in the version histories of the Zenodo record; the present corrections are of the same kind and are recorded here so that they reach readers of the PDF.

## Appendix A. Inventory of the cited targets

Every target cited in the statement and proof of [IUT3, Cor. 3.12], by the counting rule of §2: 164 occurrences, 63 targets. Unmarked items are those of [IUT3]. Roles are those of Table 2; kept targets are the first 22 rows (roles A, A+B, B). The same list is in the supplementary archive as `inventory_v120.csv`.

| target | occurrences | role |
|---|---|---|
| Def. 3.8 | 4 | definition |
| [IUT2, Cor. 4.10] | 2 | definition |
| Rem. 3.1.1 | 1 | definition |
| [IUT1, Def. 3.1] | 1 | definition |
| Def. 2.4 | 1 | definition |
| Rem. 3.9.5 | 10 | definition and premise |
| Prop. 3.9 | 4 | definition and premise |
| Thm. 3.11 | 43 | premise |
| Rem. 3.12.2 | 4 | premise |
| Rem. 3.11.1 | 5 | premise |
| Rem. 3.11.3 | 4 | premise |
| Thm. 1.5 | 4 | premise |
| Rem. 3.10.1 | 2 | premise |
| Rem. 3.9.6 | 2 | premise |
| Rem. 1.5.2 | 2 | premise |
| Rem. 2.4.2 | 1 | premise |
| Rem. 3.9.7 | 1 | premise |
| Rem. 3.8.1 | 1 | premise |
| Rem. 1.2.2 | 1 | premise |
| Rem. 1.5.4 | 2 | premise |
| Rem. 3.9.3 | 1 | premise |
| Rem. 3.9.2 | 1 | premise |
| Rem. 2.1.1 | 4 | background / comment |
| [IUT2, Rem. 3.6.4] | 3 | background / comment |
| [EtTh] (whole work) | 6 | background |

| target | occurrences | role |
|---|---|---|
| Rem. 2.3.3 | 4 | background |
| [IUT1, Ex. 5.1] | 2 | background |
| [IUT1, Def. 5.2] | 2 | background |
| Prop. 3.4 | 2 | background |
| Rem. 2.3.2 | 2 | background |
| [IUT2, Rem. 4.11.2] | 2 | background |
| [IUT2, §3] | 2 | background |
| [IUT2, Cor. 3.5] | 2 | background |
| [IUT2, Cor. 3.6] | 2 | background |
| §2 | 1 | background |
| Prop. 3.7 | 1 | background |
| Cor. 2.3 | 1 | background |
| Rem. 2.2.2 | 1 | background |
| Rem. 1.3.2 | 1 | background |
| [IUT2, Cor. 1.11] | 1 | background |
| [IUT2, Rem. 1.11.1] | 1 | background |
| [IUT2, Prop. 4.2] | 1 | background |
| [AbsTopIII], Prop. 3.2 | 1 | background |
| [AbsAnab], Prop. 1.2.1 | 1 | background |
| [IUT2, Prop. 4.4] | 1 | background |
| [IUT1, Rem. 3.4.2] | 1 | background |
| [IUT2, Rem. 1.11.3] | 1 | background |
| Rem. 1.5.1 | 1 | background |
| [IUT1, §4] | 1 | background |
| [IUT1, §5] | 1 | background |
| [IUT1, §6] | 1 | background |
| [IUT2, Rem. 4.7.6] | 2 | background |
| Rem. 3.11.2 | 1 | background |
| [IUT2, §2] | 1 | background |
| [IUT1, §2] | 1 | background |
| [IUT1, Rem. 6.12.4] | 1 | background |
| [IUT2, Rem. 4.5.3] | 1 | background |
| [IUT4] (whole work) | 1 | comment |
| [IUT4, Rem. 2.3.2] | 1 | comment |
| Rem. 3.6.2 | 1 | comment |
| Fig. 3.6 | 3 | figure |
| Fig. 3.8 | 2 | figure |
| Fig. 3.7 | 2 | figure |

## Disclosure

The proofs in this paper were found with generative AI tools (large language models) under the author's direction, and these tools were also used for the computations, the formalisation, the writing and the code that draws Figure 1. Every statement marked [P] is checked in Lean 4 with Mathlib (§11). The correspondence between the axioms and the cited results is checked against the

source texts as described in §2 and Remark 3; it is not checked by Lean. The author's own reading of the proofs is still in progress. The author takes full responsibility for the paper.

## References


[First] T. Urahigashi, *An explicit initial Θ-datum with mixed reduction at a split prime*, Zenodo, 2026. DOI 10.5281/zenodo.22537342 (all versions).

[Second] T. Urahigashi, *An initial Θ-datum over an imaginary quadratic field, and numerical values of the local quantities of inter-universal Teichmüller theory IV*, Zenodo, 2026. DOI 10.5281/zenodo.22601333 (all versions).

[Third] T. Urahigashi, *The cost of each distinction in the local computations of inter-universal Teichmüller theory IV*, Zenodo, 2026. DOI 10.5281/zenodo.22646136 (all versions).

[Fourth] T. Urahigashi, *Split bad primes and the tuple structure of inter-universal Teichmüller theory*, Zenodo, 2026. DOI 10.5281/zenodo.22648944 (all versions).

[Fifth] T. Urahigashi, *Two constants in the numerical estimates of inter-universal Teichmüller theory IV*, Zenodo, 2026. DOI 10.5281/zenodo.22659796 (all versions).

[Higuchi] J. R. Higuchi, *A formal negative result for the passage from Step (xi-e) to Step (xi-f) in IUT III*, preprint, March 31, 2026, 9 pp. DOI 10.13140/RG.2.2.10989.76006.

[Pajo] P. Pajo, *Scale-free common-container comparisons do not entail numerical inequalities: a minimum-viable no-go theorem motivated by the IUT III Theorem 3.11-to-Corollary 3.12 bridge*, preliminary research note, April 26, 2026, 14 pp. DOI 10.13140/RG.2.2.33620.16005.

[EtTh] S. Mochizuki, *The étale theta function and its Frobenioid-theoretic manifestations*, Publ. Res. Inst. Math. Sci. **45** (2009), 227–349.

[IUT1] S. Mochizuki, *Inter-universal Teichmüller theory I*, Publ. Res. Inst. Math. Sci. **57** (2021), 3–207. Consulted author version: May 2020.

[IUT2] S. Mochizuki, *Inter-universal Teichmüller theory II*, Publ. Res. Inst. Math. Sci. **57** (2021), 209–401. Consulted author version: December 2020.

[IUT3] S. Mochizuki, *Inter-universal Teichmüller theory III*, Publ. Res. Inst. Math. Sci. **57** (2021), 403–626. Consulted author version: May 2020.

[IUT4] S. Mochizuki, *Inter-universal Teichmüller theory IV*, Publ. Res. Inst. Math. Sci. **57** (2021), 627–723. Consulted author version: April 2020.

[Rpt] S. Mochizuki, *Report on discussions, held during the period March 15–20, 2018, concerning inter-universal Teichmüller theory*, RIMS, Kyoto University, February 2019 version, 45 pp.

[Cmt18a] S. Mochizuki, *Comments on the manuscript by Scholze–Stix concerning inter-universal Teichmüller theory (IUTch)*, July 2018, 8 pp.

[Cmt18b] S. Mochizuki, *Comments on the manuscript (2018-08 version) by Scholze–Stix concerning inter-universal Teichmüller theory (IUTch)*, September 2018, 5 pp.

[Form] S. Mochizuki, *On the formalization of IUT: a preliminary progress report* (joint work in progress with Y. Hoshi, G. Yamashita, Y. Yang), slides, April 2026, 14 pp.

[LANA] Project LANA, *Project LANA interim report on IUT theory*, July 2026, 50 pp.

[JoshiIII] K. Joshi, *Construction of arithmetic Teichmüller spaces III: A 'Rosetta Stone' and a proof of Mochizuki's Corollary 3.12*, arXiv:2401.13508.

[JoshiIV] K. Joshi, *Construction of arithmetic Teichmüller spaces IV: Proof of the abc-conjecture*, arXiv:2403.10430v2, February 2025.

[JoshiLANA] K. Joshi, *Comments on the LANA Project Report of Kato et al.*, July 30, 2026, 6 pp.

[SS] P. Scholze, J. Stix, *Why abc is still a conjecture*, manuscript, July 16, 2018. Footnote numbers refer to the version dated August 23, 2018.

[lana] Project LANA, *lana-agents/iut*, Lean repository, `https://github.com/lana-agents/iut`; file `Iut/WallEquiv/NonDerivable.lean`, added on 13 October 2026 (commit 180c049), consulted at commit 553076a (3 November 2026).

[Ri] D. Richards, *motizukipdf — Mochizuki's papers, plus a toolkit for the data types of IUT*, GitHub repository `https://github.com/oneman/motizukipdf`; Lean port `lean/IUT/LeanForm.lean` at commit 2665db1 (30 October 2026).

[RF] T. Urahigashi, *No reading avoids the height inequality: 36 readings of Corollary 3.12 and the conservation of the height load*, version 1.0.0, Zenodo, 2026. DOI 10.5281/zenodo.23965952.

[SL] T. Urahigashi, *Segment or loop? A homotopy-type-theoretic overview of the dispute over Corollary 3.12 of inter-universal Teichmüller theory*, version 1.3.0, Zenodo, 2026. DOI 10.5281/zenodo.23991545.

[TC] T. Urahigashi, *Two computations of the height inequality and the multiradial algorithm: reading IUT from the viewpoint of computer science*, version 1.0.0, Zenodo, 2026. DOI 10.5281/zenodo.23968299.

[II] T. Urahigashi, *Reading the IUT dispute through indexed identifications: an educational model in undergraduate mathematics*, Zenodo, 2026. DOI 10.5281/zenodo.23931345 (all versions).

[Takkun] Takkun-kohinata, *IUT_LEAN*, Lean repository, `https://github.com/Takkun-kohinata/IUT_LEAN`, commit 9d56c46 (15 July 2026); files `IutLean/Multiradial/ThetaHullDichotomy.lean` and `DisputedCore.lean`.

[Saito] Y. Saito, *A registered-provenance audit of IUT III, Corollary 3.12, Step (xi)*, version 2.0.0, Zenodo, 26 July 2026. DOI 10.5281/zenodo.21596198.

[sfingali] sfingali, *The Scholze–Stix objection, formally: the arithmetic is agreed, the q-side is absent*, Zenodo, 2 August 2026. DOI 10.5281/zenodo.21757640.

[Zeng] C. Zeng, *An integral-structure inflation in the hull operation of IUT Corollary 3.12*, Zenodo, first version 29 October 2026. DOI 10.5281/zenodo.23754276 (all versions).

[Geiger] L. Geiger, *The bridge certificate: a line-level audit instrument for inter-theoretic transfer arguments*, Zenodo, first version 14 August 2026. DOI 10.5281/zenodo.21925803 (all versions).

[BS] T. Urahigashi, *Bridge and size: the Kodaira–Spencer morphism, arithmetic derivatives and Step (xi) of IUT as lax 2-cells*, version 1.0.0, Zenodo, 2026. DOI 10.5281/zenodo.23984201.